VIETNAM GENERAL CONFEDERATION OF LABOR

TON DUC THANG UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY

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ANALYZING THEMES IN VIETNAMESE SONG LYRICS

INFORMATION TECHNOLOGY PROJECT

COMPUTER SCIENCE

HO CHI MINH CITY, 2025

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MSc. Nguyen Quoc Binh

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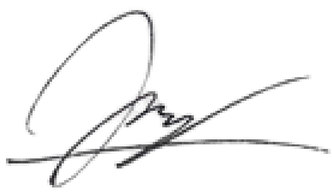
Ho Chi Minh City, November 16, 2025.

Author

(Sign and specify full name)

Trinh Minh Tich Thien



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Trinh Minh Tich Thien *Do Hoang Duy*

ANALYZING THEMES IN VIETNAMESE SONG LYRICS

ABSTRACT

This research conducts the large-scale quantitative analysis of Vietnamese music history, analyzing a corpus of **61,426 songs** (1900–2025) to map the nation's cultural evolution. By benchmarking five distinct topic modeling architectures—LSA, NMF, LDA, BERTopic, and CTM—we identified the **Contextualized Topic Model (CTM)** as the optimal approach, achieving the highest Coherence (0.6385) and Diversity (0.7740) scores. The resulting **Quantitative Topic Timeline** statistically confirms a structural shift from the "Collective Survival" narratives of the Resistance Era to the "Individual Expression" of the modern age. Furthermore, the analysis highlights distinct micro-trends, including the therapeutic "Sadness Paradox", the post-2018 surge in Rap, and the enduring resilience of traditional themes like Tet. These findings demonstrate that computational methods can objectively reconstruct historical narratives, establishing a robust data-driven framework for Vietnamese cultural studies.

Keywords: *Computational Musicology, Topic Modeling, CTM, Vietnamese History, Natural Language Processing.*

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ABBREVIATIONS

BERT	Bidirectional Encoder Representations from Transformers
BERTopic	BERT-based Topic Modeling
BoW	Bag-of-Words
CombinedTM	Combined Topic Model
c-TF-IDF	Class-based Term Frequency-Inverse Document Frequency
CTM	Contextualized Topic Models
GPU	Graphics Processing Unit
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
LDA	Latent Dirichlet Allocation
LSA	Latent Semantic Analysis
NER	Named Entity Recognition
NLP	Natural Language Processing
NMF	Non-negative Matrix Factorization
NPMI	Normalized Pointwise Mutual Information
POS	Part-of-Speech
SBERT	Sentence-BERT
SVD	Singular Value Decomposition
TF-IDF	Term Frequency-Inverse Document Frequency
UMAP	Uniform Manifold Approximation and Projection
VSM	Vector Space Model

MATH SYMBOLS

V^T	Document-Topic Matrix (LSA)
e	Contextualized Embedding Vector (BERT)
H	Factorized Coefficient/Encoding Matrix (NMF)
K	Number of Topics
M	Total Number of Documents (Corpus Size)
N	Total Number of Words in a Document
U	Orthogonal Term-Topic Matrix (LSA)
W	Factorized Feature/Basis Matrix (NMF)
w	Observed Word
X	Term-Document Matrix
z	Latent Topic Assignment
Σ	Diagonal Matrix of Singular Values (LSA)
α	Dirichlet Prior (Document-Topic Density)
β	Dirichlet Prior (Topic-Word Density)
θ	Per-document Topic Distribution
μ	Mean of Latent Topic Distribution (CombinedTM)

CHAPTER 1: INTRODUCTION

1.1. Rationale for Choosing the Topic

Popular music is not merely a form of entertainment. According to contemporary computational musicology, it serves as a vital cultural product that reflects the social, psychological, and cultural context of its era (Hunke, Huber, & Steffens, 2025). In the specific case of Vietnam, the evolution of song lyrics provides a crucial historical lens. These lyrics trace the nation's journey through clearly defined epochs: starting with the romantic introspection of the **Pre-War period** (*Nhạc Tiền Chiến*), moving to the mobilized patriotism of the **Resistance Wars** (*Nhạc Đỏ*), and finally evolving into the pluralistic individualism of the **Renovation era** (*Đổi Mới*). Each period features a distinct vocabulary and thematic focus, effectively functioning as an archive of the "spirit of the times".

Preserving Vietnam's cultural history requires an understanding of these lyrical shifts. Lyrics act as a collective diary for the Vietnamese people, documenting how core values—such as patriotism, filial piety, and romantic love—have changed over the last 70 years. Analyzing this evolution provides insight into the changing Vietnamese identity. Specifically, it tests the hypothesis that the nation has shifted from the collective sacrifice of the 20th century to the personal expression of the 21st century.

Despite the sociological importance of these changes, current research on Vietnamese music history relies heavily on qualitative methods. Historians and critics typically define these eras using subjective interpretation, emotional resonance, and isolated case studies. While useful, this approach lacks a rigorous quantitative foundation. Common academic claims—such as "Pre-war music is highly literary" or "Modern music has shifted entirely from political to personal themes"—remain descriptive observations rather than statistically verified facts. Additionally, existing quantitative studies focus almost exclusively on Western corpora, such as English and German. No systematic effort has been made to apply large-scale topic modeling to the Vietnamese corpus to determine if the nation follows global trends or retains a unique cultural path.

The recent availability of large-scale digital music archives offers a timely chance to address this limitation. Researchers can now aggregate and process tens of thousands of lyrics, applying **Natural Language Processing (NLP)** and **Text Mining** techniques to musicology. This technological advancement allows for a shift from the “close reading” of individual tracks to the “distant reading” of an entire cultural corpus. This approach is necessary to map complex societal dynamics over time that are invisible to the human ear (Mauch, MacCallum, Levy, & Leroi, 2015).

Consequently, a systematic study is needed to analyze this massive dataset using historical-quantitative methods. This research aims to bridge the gap between cultural history and data science. By using advanced computational models, we seek to re-evaluate traditional academic judgments with clear numerical evidence. This reduces subjective bias and provides a comprehensive, objective map of how Vietnamese song lyrics have evolved.

1.2. Related Work

Computational analysis of Vietnamese lyrics is still in its early stages, but the global field of computational musicology is well-established. It provides a rigorous methodological baseline for this study. Existing scholarship focuses mostly on Western corpora (English and German), but it demonstrates the power of NLP to uncover long-term cultural dynamics:

- **Evolutionary Trends:** A seminal analysis of US Billboard charts from 1960 to 2010 used signal processing and text mining to show that musical evolution is defined by sudden “revolutions” driven by the introduction of new genres rather than gradual change. This finding sets a critical precedent for using computational methods to track historical shifts in cultural production (Mauch, MacCallum, Levy, & Leroi, 2015).

- **Emotional Shifts:** Quantitative research has challenged the assumption that popular music is inherently "happy." By examining five decades of American lyrics, researchers identified a distinct long-term trend toward "sadder" and more emotionally ambiguous content. This shift is attributed to the rise of individualism, where listeners increasingly prefer music that mirrors personal complexity rather than collective joy (Schellenberg & von Scheve, 2012).
- **The "German Study" Benchmark:** The most relevant work to our methodology is a recent study applying Latent Dirichlet Allocation (LDA) to German lyrics from 1954 to 2022. By successfully mapping the decline of "Post-war longing" and the simultaneous rise of "Status & Society," this study proves that topic modeling can accurately reconstruct a nation's historical trajectory through its songbook (Hunke, Huber, & Steffens, 2025).

1.3. Research Aim

Previous studies often relied on single algorithms, which is a methodological gap. To address this, our research proposes a **comparative benchmarking framework**. We will train and evaluate five distinct topic modeling architectures to find the mathematically optimal approach for the Vietnamese language:

- **Latent Dirichlet Allocation (LDA):** The industry-standard generative probabilistic model utilized in the majority of existing computational musicology literature to identify latent topic mixtures (Blei, Ng, & Jordan, 2003).
- **Non-negative Matrix Factorization (NMF):** An additive parts-based model known for producing sharper, more distinct topics by enforcing strict non-negativity constraints on the factorized matrices (Lee & Seung, 1999).

- **BERTopic:** A state-of-the-art neural model utilizing pre-trained transformer embeddings (specifically multilingual models) to capture semantic context beyond simple word counts (Grootendorst, 2022).
- **Latent Semantic Analysis (LSA):** A foundational technique utilizing Singular Value Decomposition (SVD) to identify baseline semantic patterns through linear algebra (Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990).
- **Contextualized Topic Models (CombinedTM):** A hybrid neural architecture that integrates pre-trained representations with a probabilistic framework, leveraging embeddings to resolve semantic ambiguity while maintaining interpretability (Bianchi, Terragni, & Hovy, 2021).

We will evaluate these models based on **Coherence Score** and **Topic Diversity** metrics to identify the single "Champion Model." This best-performing model will then be used to construct a **Quantitative History of Vietnamese Music**, visualizing the rise and fall of prominent themes across decades to verify historical hypotheses.

1.4. Object and Scope of Research

The primary object of this study is the **semantic structure of Vietnamese popular music lyrics**. Unlike Music Information Retrieval (MIR) studies that focus on audio features like pitch, rhythm, or timbre, this research treats song lyrics as a textual corpus. We analyze them as cultural documents that encode social values and historical narratives.

To ensure the feasibility and reliability of the analysis, the research scope is strictly defined across three specific dimensions. First, regarding **Data Scope**, the study analyzes a comprehensive dataset of approximately 60,000 songs collected from major digital archives, spanning the historical timeline from the Pre-War period (Pre-1954) through the Resistance Wars and Subsidy Era, extending to the contemporary Renovation (*Đổi Mới*) and Modern era. Second, in terms of **Linguistic Scope**, the analysis is strictly limited to Modern Vietnamese; songs containing significant portions of foreign languages (English, Chinese, French) or dialect-heavy folk variants are excluded to ensure model stability. Finally, the **Methodological Scope** is restricted exclusively to unsupervised topic modeling using the five distinct architectures listed in the Research Aim. Other Natural Language Processing tasks, such as sentiment analysis or supervised genre classification, fall outside the boundary of this thesis.

CHAPTER 2: THEORETICAL FOUNDATION

2.1. Vietnamese Computational Linguistics

Analyzing the historical evolution of Vietnamese lyrics requires addressing the specific linguistic features that set Vietnamese apart from Western languages used in earlier musicology studies. From a computational standpoint, the main obstacle is the **“Word Boundary Problem”**. In English, whitespace reliably separates semantic units. Vietnamese, however, is an isolating, monosyllabic language where whitespace separates syllables, but not necessarily words.

In Vietnamese, meaning is frequently constructed through compound words composed of two or more syllables; for instance, the syllable *đất* denotes “earth” and *nước* denotes “water,” but the compound *đất nước* signifies “nation” or “country”. A standard “Bag-of-Words” model relying on whitespace tokenization would treat these as separate, unrelated entities, effectively stripping the text of its specific patriotic meaning. Consequently, the theoretical prerequisite for this study is **Word Segmentation**, a process that groups syllables into meaningful compound tokens to define the correct unit of analysis before any matrix factorization occurs (Nguyen, Nguyen, Vu, Dras, & Johnson, 2018).

2.2. The Algebraic Paradigm: Text as Geometry

The foundational method for extracting meaning from text involves **Vector Space Models (VSM)**, which translate linguistic differences into mathematical vectors. This study uses two distinct linear algebra techniques to uncover latent structures within the high-dimensional document-term matrix.

Latent Semantic Analysis (LSA) serves as the baseline for topic modeling. It is based on the “Distributional Hypothesis”, which argues that words with similar meanings inevitably appear in similar contexts (Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990). Theoretically, LSA assumes that the raw term-document matrix is “noisy” due to synonyms and polysemy. To resolve this, the core mechanism of LSA is **Singular Value Decomposition (SVD)**.

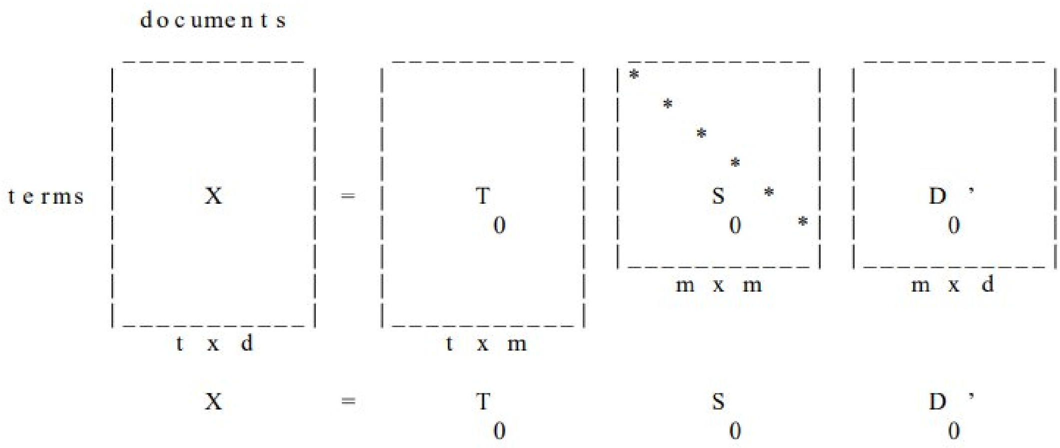

Figure 2.1: Singular Value Decomposition (SVD) of the Term-Document Matrix. The matrix X is decomposed into Term vectors (T_0), Singular values (S_0), and Document vectors (D'_0).

Source: (Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990)

The model decomposes the term-document matrix X into three component matrices: $X \approx U \Sigma V^T$. By truncating the singular values in Σ to capture only the most significant variance, the model projects data into a lower-dimensional “semantic space” that filters out noise (Landauer, Foltz, & Laham, 1998). In this projection, the model detects thematic similarities geometrically. For example, it can recognize that two songs are both about “Sadness” even if one uses the word “cry” and the other uses “weep.” However, LSA has a major theoretical limitation: its vector dimensions can contain negative values. In the context of lyrics, this creates interpretability problems—mathematically, a song cannot contain a “negative” amount of a topic like “Love”—which renders the results abstract.

To solve these interpretability constraints, Non-negative Matrix Factorization (NMF) (Lee & Seung, 1999) distinguishes itself through a strict “non-negativity constraint", forcing all values in the factorized feature (W) and coefficient (H) matrices to be greater than or equal to zero: $V \approx WH$ subject to $W \geq 0, H \geq 0$

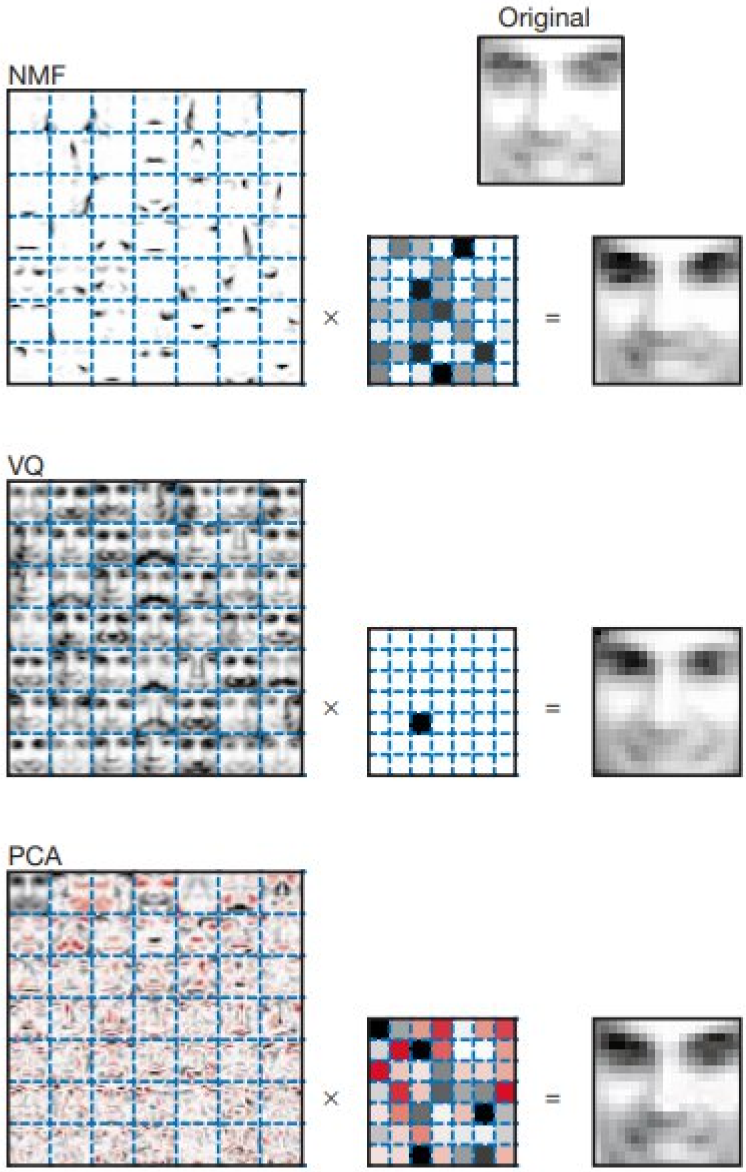

Figure 2.2: Comparison of NMF, VQ, and PCA representations. The top row (NMF) demonstrates the learning of localized "parts" (eyes, noses), contrasting with the holistic "eigenfaces" found in PCA (bottom row), illustrating the interpretability advantage of non-negative constraints.

Source: *(Lee & Seung, 1999)*

This constraint fundamentally alters the nature of the representation. While LSA is holistic, NMF operates on an **"additive parts-based"** assumption (Lee &

Seung, 1999). This logic is intuitive and well-suited for lyrics: a song about "War" is constructed by adding words like "soldier" and "gun" to a baseline; one cannot "subtract" meaning. Because of this mathematical constraint, the model must learn sparse, additive parts that can be superimposed to recreate the original lyrics. This property typically produces sharper, more distinguishable topics compared to unconstrained factorization.

2.3. The Probabilistic Paradigm: Text as Statistics

Unlike linear algebra models that treat text as geometric vectors, probabilistic models view text as random distributions generated by specific statistical processes. **Latent Dirichlet Allocation (LDA)** is the industry standard for topic modeling in computational musicology. It represents a significant theoretical improvement over "hard" clustering algorithms like K-Means (Blei, Ng, & Jordan, 2003).

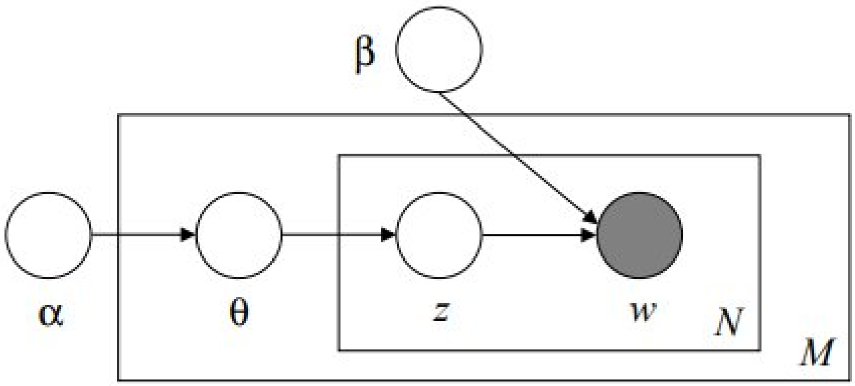

Figure 2.3: Graphical Model (Plate Notation) of Latent Dirichlet Allocation. The outer box represents the documents (M), while the inner box represents the repeated choice of topics (z) and words (w) within a document (N).

Source: *(Blei, Ng, & Jordan, 2003)*

In the nuanced context of cultural data, a song is rarely a monolith. Unlike clustering, which forces a document into a single category, LDA models **mixed membership**. For instance, a single track describing a soldier's letter home can function simultaneously as a "War" song, a "Family" song, and a "Love" song. LDA captures this semantic complexity by proposing a **"generative story"** for document creation. It assumes every document is a finite mixture over an underlying set of topics, and each topic is, in turn, an infinite mixture over a set of word probabilities.

$$P(w \mid z) \sim \text{Dirichlet}(\beta)$$

$$P(z \mid \theta) \sim \text{Dirichlet}(\alpha)$$

This probabilistic framework allows us to map the emotional landscape of Vietnamese music with high fidelity. It acknowledges that themes naturally overlap—for instance, the topic of "Spring" (*Xuân*) frequently intersects with "Love" (*Tình yêu*) in the Vietnamese musical tradition. However, a major theoretical limitation of LDA is the "independence assumption"—the Dirichlet prior assumes that the presence of one topic has no effect on the presence of another.

2.4. The Neural Paradigm: Text as Semantic Meaning

The field has recently evolved beyond simple word counting by using the **"Attention Mechanism"** of deep learning to capture semantic context. Traditional models like LDA and NMF rely on the "Bag-of-Words" assumption, which ignores word order. In this framework, the Vietnamese word *bạc* has the same numerical value whether it refers to the color "silver" or the concept of "ungrateful" (*bạc bẽo*).

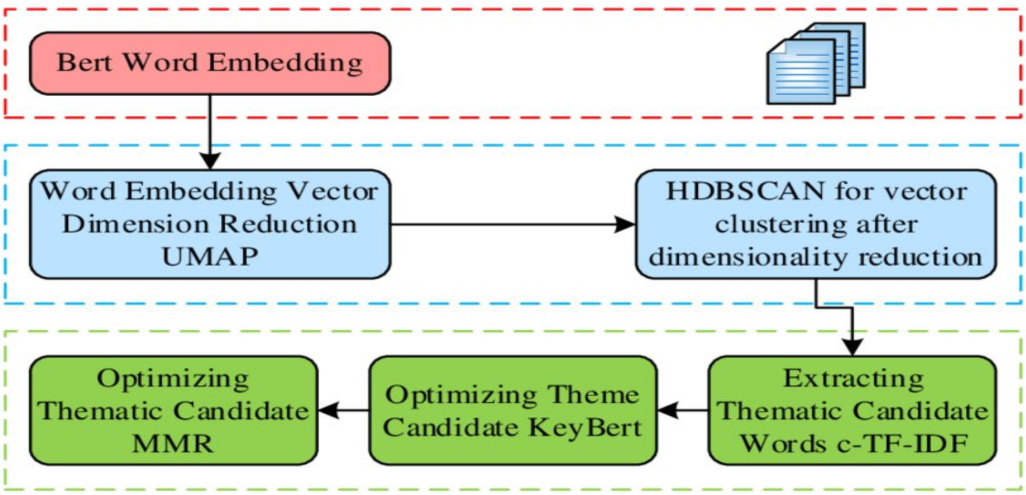

Figure 2.4: The Architecture Pipeline of BERTopic. The process follows a modular structure: (1) Embedding generation via BERT, (2) Dimensionality reduction (UMAP) and Clustering (HDBSCAN), and (3) Topic Representation using c-TF-IDF and KeyBERT optimizations.

Source: *(Lu, et al., 2024)*

BERTopic (Grootendorst, 2022) resolves this by using Transformer architectures to generate **Contextual Embeddings**. These models map words into a dense vector space where position and surrounding context determine the vector's value, effectively capturing polysemy. Methodologically, BERTopic changes the analytical paradigm: it approaches topic discovery not as a probabilistic mixture problem, but as a **Clustering Task**. The underlying assumption is that documents (songs) sharing similar semantic themes will cluster together in a high-dimensional vector space. A key theoretical innovation is the **Class-based TF-IDF (c-TF-IDF)** procedure, which treats all documents in a cluster as a single "meta-document" to extract class-specific keywords. This allows the model to identify topics based on semantic meaning rather than just lexical overlap.

2.5. The Hybrid Paradigm

To resolve the limitations of both statistical independence (LDA) and the lack of interpretability in pure clustering, we utilize the **Contextualized Topic Model (CombinedTM)**. While the statistical concept of correlating topics was originally introduced via the Correlated Topic Model (Blei & Lafferty, Correlated Topic Models, 2006), our research utilizes the neural architecture proposed by (Bianchi, Terragni, & Hovy, 2021). This approach represents a significant evolution in unsupervised learning by integrating pre-trained contextualized representations directly into the generative process.

Traditional probabilistic models, such as Latent Dirichlet Allocation (LDA), treat documents as exchangeable "bags of words", effectively discarding the syntactic and semantic structure of sentences (Blei, Ng, & Jordan, 2003). Consequently, these models often struggle to resolve linguistic ambiguity, particularly in short texts or languages with high polysemy. The CombinedTM overcomes this deficiency by operating as a **hybrid architecture** within the framework of Neural Variational

Inference, specifically extending the Autoencoded Variational Inference for Topic Models (AVITM) (Srivastava & Sutton, 2017).

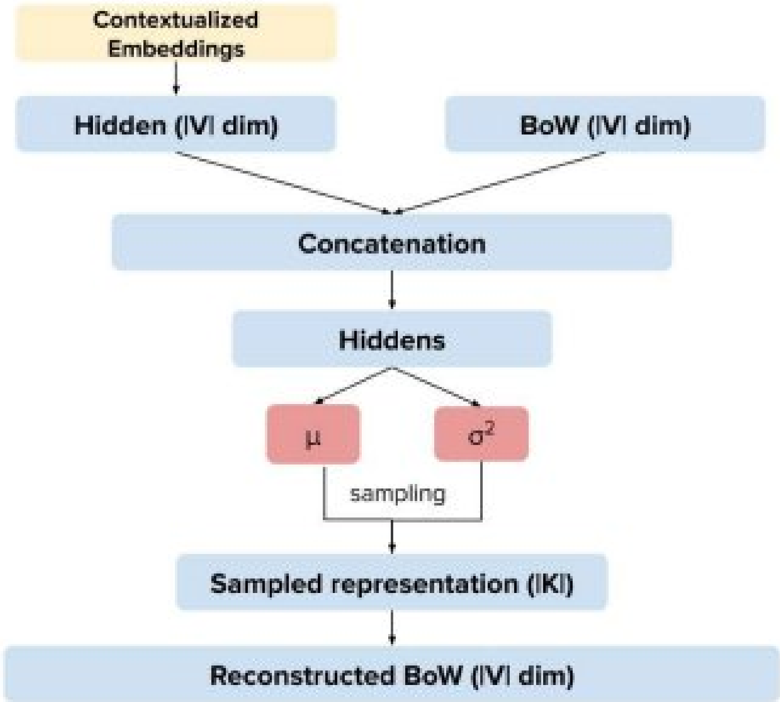

Figure 2.5: Schematic view of the CombinedTM Architecture. The inference network (top) utilizes a concatenation strategy, combining Contextualized Embeddings (BERT) with Bag-of-Words (BoW) to estimate the latent distribution parameters (μ,Σ).

Source: *(Bianchi, Terragni, & Hovy, 2021)*

The core innovation of the CombinedTM lies in its "concatenation" strategy within the inference network (encoder). In traditional Neural Topic Models (like ProdLDA), the encoder maps a simple Bag-of-Words vector to a latent topic distribution. The CombinedTM, however, modifies this input stage to utilize complementary information:

1. **The LDA Component (Bag-of-Words, x):** It retains the frequency-based vector to ensure the model pays attention to the explicit words present in the document.
2. **The BERT Component (Contextualized Embedding, e):** It integrates a dense vector generated by a pre-trained transformer.

Mathematically, the encoder takes the combined representation $x \oplus e$ (concatenation) and passes it through a neural network to estimate the parameters (μ, Σ) of the latent topic distribution. As noted by Bianchi et al., this allows the model to "account for the semantic meaning of the text" via e, while the decoder (generative network) focuses on reconstructing the Bag-of-Words x. This duality allows the model to resolve ambiguity—for instance, distinguishing "thương" as *pity* versus *love* based on the surrounding sentence structure provided by BERT—while ensuring the final topics remain interpretable lists of words, similar to LDA.

CHAPTER 3: METHODOLOGY

This study established a data processing pipeline designed to convert unstructured text into measurable historical data. The methodology follows four sequential stages: (1) Data Acquisition and Linguistic Preprocessing, tailored to the Vietnamese language; (2) The implementation and parameter optimization of five specific topic modeling architectures; (3) A comparative evaluation using semantic coherence and diversity metrics to select the optimal model for mapping the evolutionary history of Vietnamese music.

3.1. Dataset

The research analyzes a corpus of **61,525 songs**, aggregated from five digital archives to cover the spectrum of Vietnamese popular music from the mid-20th century to the present. Domestic content dominates the dataset, comprising 96.2% (59,173 songs) of the total, with the remainder consisting primarily of foreign melodies adapted with Vietnamese lyrics (*Nhạc ngoại lời Việt*). A multi-label system categorizes these tracks into various genres. As illustrated in Figure 3.1, the distribution leans toward modern mainstream genres, specifically **"Youth Music"** (*Nhạc Trẻ* – 22,042 songs) and **"Lyrical Ballads"** (*Ballad/Trữ Tình* – 10,307 songs). However, the dataset preserves historically significant categories such as **"Golden Music"** (*Nhạc Vàng*), **"Revolutionary Music"** (*Nhạc Đỏ*), and **"Trinh Music"** (*Nhạc Trịnh*), which facilitates the isolation of vocabularies specific to these movements. Recent linguistic shifts are captured through genres like **"Rap/Hip-hop"** (2,769 songs) and **"Indie"**.

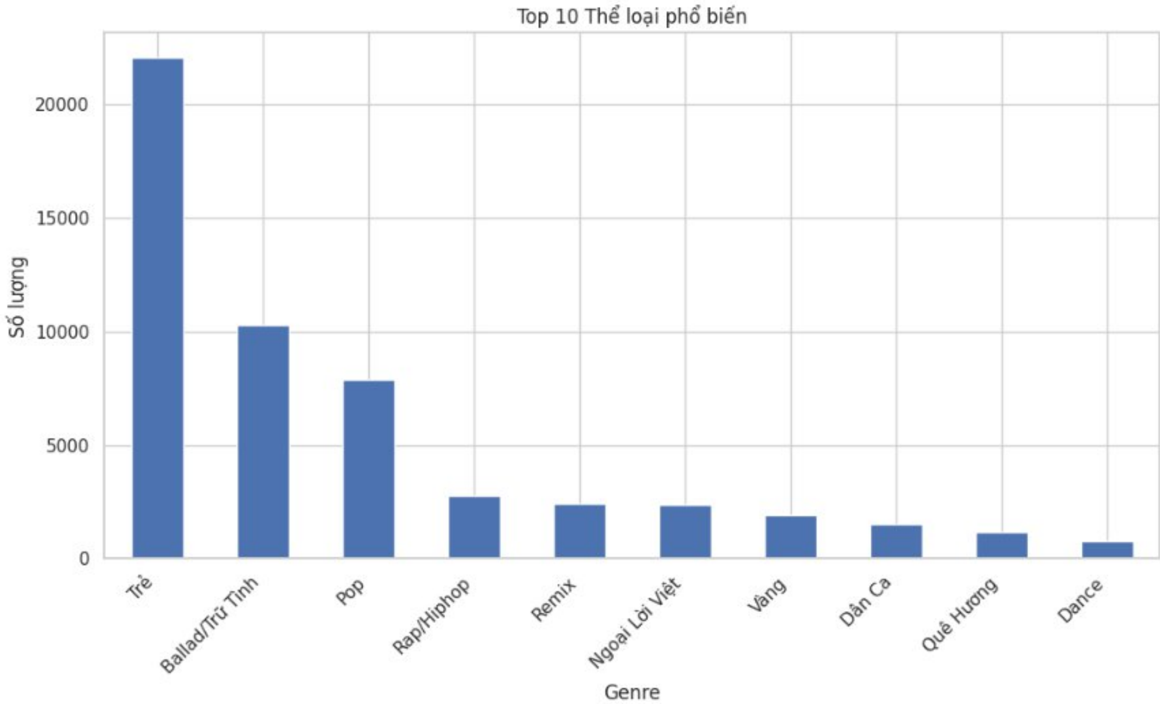

Figure 3.1: *Top 10 Genre Distribution*

A primary goal of this study is tracking lyrical evolution over time. Since web-scraped metadata is often incomplete, we applied a data imputation strategy. Release years provided by source websites were prioritized, followed by an automated enrichment process using APIs from Spotify, iTunes, and MusicBrainz. A traceability log (e.g., tagged as “filled ‘year’ via Spotify”) was maintained for transparency. Despite this, a portion of the dataset—particularly pre-digital era music—lacks digital fingerprints. The impact of this “Recency Bias” is addressed in the Experimental Data analysis (Section 4.1).

To verify data quality, we conducted a missing value assessment (Figure 3.2). The critical features for modeling—lyrics and titles—were 100% complete. Additionally, the dataset identifies over 12,000 unique composers and 13,000 unique lyricists. This authorial diversity suggests the models will reflect a collective cultural lexicon rather than the styles of a few prolific individuals.

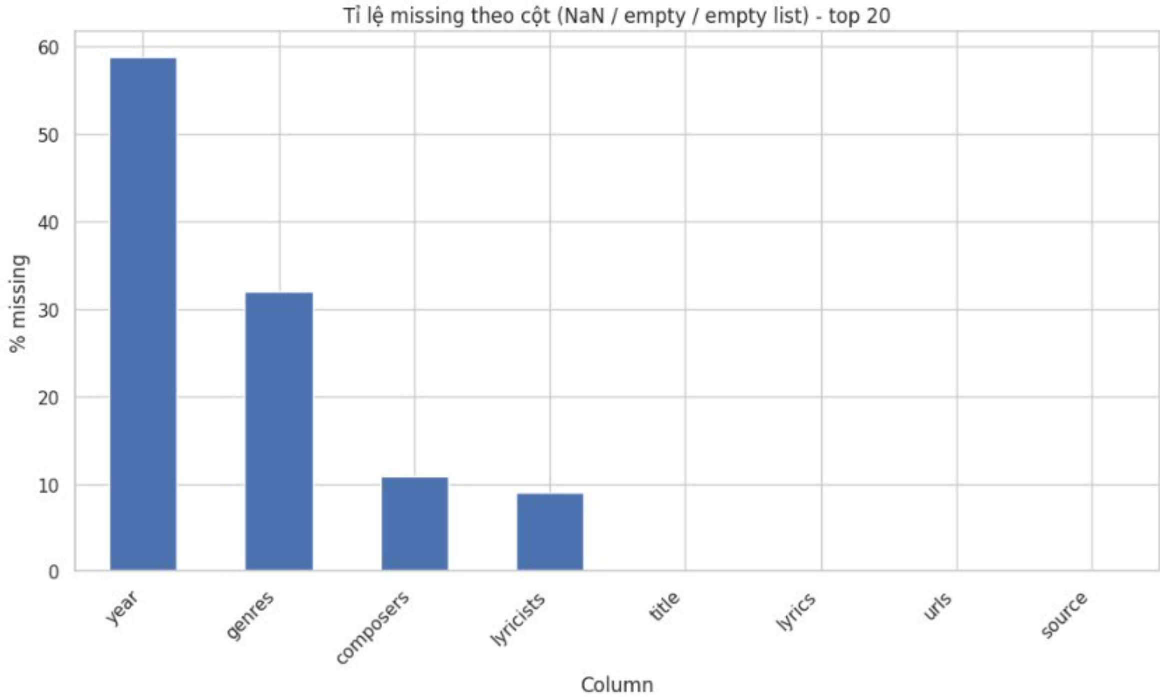

Figure 3.2: Missing Value Rates by Column

Data Preprocessing: Raw lyrics contain formatting inconsistencies, errors, and complex structures that require standardization. Before modeling, the data passed through a four-phase cleaning pipeline.

First, we addressed **Structural Cleaning and Normalization**. We used strict Regular Expressions (Regex) to remove non-lyrical content. Text within brackets **[...]** and parentheses **(...)** often containing chords or singer notes—was stripped. Digital artifacts like URLs, emails, HTML tags, and punctuation were removed. Finally, all text was converted to the **Unicode NFKC standard** to ensure consistent encoding.

Next, to address the Vietnamese “Word Boundary” problem, where meaning relies on compound words, we integrated the **VnCoreNLP** toolkit for segmentation, Part-of-Speech (POS) tagging, and Named Entity Recognition (NER). To handle memory constraints, we engineered a “Smart Chunking” function (*split_text_smart*) that divides lyrics into segments under 150 words based on line breaks, processes them, and reassembles them. This ensured compound words like *đất_nước* were treated as single tokens.

Subsequently, we applied **Statistical Phrasing** to capture slang or poetic combinations often missed by standard dictionaries. Using a Bigram model trained on the entire corpus with a strict threshold score of 100, the system automatically merged words that co-occurred significantly more often than chance (e.g., merging *người* and *dưng* into the compound *người_dưng*). We then aligned these statistical bigrams with the grammatical tags from the previous stage, combining grammatical accuracy with the flexibility of context-aware phrasing.

Finally, the data was passed through a **"Semantic Sieve"** to filter out noise. We removed standard stopwords and applied a strict POS Filter, retaining only "Content Words"—specifically **Nouns (N)**, **Verbs (V)**, and **Adjectives (A)**. Crucially, to mitigate potential tagger errors, we implemented a "Safety Net" using **NER Override**: if the model identified a word as a Location (LOC) or Organization (ORG), it was preserved regardless of its POS tag, ensuring historical markers like "Hanoi" or "Saigon" were not lost. To further refine the vocabulary, we pruned tokens appearing in fewer than 20 documents or more than 70% of the corpus, and discarded any songs containing fewer than 5 valid tokens.

3.2. Latent Dirichlet Allocation (LDA)

We implemented the probabilistic framework using the Python *gensim* library. Hyperparameters were tailored to the Vietnamese corpus to ensure meaningful patterns. Before training, the vocabulary was optimized using *filter_extremes*. We removed words appearing in fewer than 30 documents to reduce outliers and, significantly, discarded words appearing in more than 40% of the corpus. This addresses **Zipf's Law** by filtering out generic high-frequency terms, forcing the model to focus on the discriminative "middle-frequency" vocabulary (Hunke, Huber, & Steffens, 2025).

Furthermore, a critical adjustment in our model configuration was setting the Dirichlet prior to **alpha='asymmetric'**. While default LDA implementations often utilize a symmetric alpha (assuming a document is a uniform blend of all topics), our asymmetric setting reflects the reality of pop music: songs are usually focused on one or two dominant themes rather than a vague spread of everything. By enforcing asymmetry, we encouraged the model to discover sparse distributions, theoretically resulting in sharper, more interpretable topics. We also set **passes=10**, allowing the model to iterate over the entire corpus ten times to ensure the probability distributions converged to a stable posterior state before evaluation.

3.3. Non-negative Matrix Factorization (NMF)

To ensure the robustness of our thematic discovery, we did not rely solely on the probabilistic approach of LDA. We employed **Non-negative Matrix Factorization (NMF)** as a complementary analytical model. While LDA relies on raw frequency counts (Bag-of-Words), NMF performs best when the input matrix reflects the importance of words rather than just their occurrence. Therefore, we transformed our corpus using **Term Frequency-Inverse Document Frequency (TF-IDF)** before feeding it into the model. This statistical weighting acts as a "significance filter." The TF component captures how often a word appears in a specific song, while the IDF component penalizes words that appear ubiquitously across the entire corpus (such as generic pop fillers like *yêu* or *em*). This weighting is crucial for music analysis because it amplifies the signal of unique, genre-defining vocabulary while suppressing background noise.

To ensure stability and reproducibility, we aligned the vocabulary pruning strategy with the LDA model, removing terms appearing in fewer than 20 documents or more than 40% of the corpus. Furthermore, we configured the iterative solver to run for ten passes (*passes=10*) with a strict convergence threshold. These settings ensure that the matrix factorization process avoids local minima and converges to a stable global solution, providing a fair and rigorous comparison against the probabilistic baseline.

3.4. BERTopic (Neural Topic Modeling)

To complement our traditional statistical models (LDA and NMF), we implemented **BERTopic** using a rigorous three-stage pipeline designed to transform raw text into meaningful semantic clusters. First, we established **Contextual Embedding** using the Sentence-BERT (SBERT) framework. We specifically utilized the *paraphrase-multilingual-MiniLM-L12-v2* model. This pre-trained transformer is optimized for multilingual tasks, ensuring robust handling of Vietnamese linguistic nuances. To manage the computational load of the extensive corpus, we implemented batch processing to generate high-fidelity embeddings without compromising memory efficiency.

Next, we addressed **Dimensionality Reduction and Clustering**. Since the raw embedding vectors exist in a massive multi-dimensional space (384 dimensions)—which is too sparse for effective clustering due to the "curse of dimensionality"—we employed **UMAP** (Uniform Manifold Approximation and Projection). We configured the algorithm to compress these vectors down to 5 dimensions, setting *n_neighbors=15* and *metric='cosine'* to preserve the local semantic structure. To group the songs, we employed **HDBSCAN**. Unlike K-Means, which forces every data point into a cluster, HDBSCAN is density-based. It identifies core clusters of songs and, crucially, classifies "outliers" (songs that do not fit into any coherent theme) as noise (Topic -1), preventing our topics from being contaminated by irrelevant data.

Finally, we refined the **Topic Representation**. Once clusters were identified, the model required accurate labeling. While the standard implementation uses class-based TF-IDF (c-TF-IDF), we enhanced the representation using the **KeyBERTInspired** model. This technique extracts keywords that are most cosine-similar to the topic's centroid embedding, ensuring that the resulting labels are semantically representative of the cluster's core meaning rather than simply identifying the most frequent words.

3.5. Latent Semantic Analysis (LSA)

To benchmark the performance of our probabilistic (LDA) and neural (BERTopic) models, we compared them against **Latent Semantic Analysis (LSA)** using the *TruncatedSVD* algorithm, the foundational technique in computational semantics. A critical limitation of LSA is its sensitivity to high-frequency words; since SVD attempts to maximize variance, it often latches onto the most frequent words in the corpus, creating generic “super-topics”.

To counteract this, we implemented a highly specific feature engineering pipeline using the *TfidfVectorizer* with an **aggressive vocabulary pruning strategy**. Most notably, we utilized a strict maximum document frequency threshold (*max_df=0.15*), explicitly discarding any word appearing in more than 15% of the corpus. Unlike standard NLP tasks that typically use a relaxed threshold (e.g., 80%), this restrictive filter was necessary to strip away not just grammatical stopwords but also common genre tropes (such as *anh*, *em*, *yêu*). This forces the SVD algorithm to find patterns in the “long tail” of the vocabulary. Additionally, we enforced a minimum frequency filter (*min_df=10*) to ensure that the resulting topics were based on shared cultural vocabulary rather than idiosyncratic typos or rare artifacts.

3.6. Contextualized Topic Models (CTM)

The model was implemented utilizing the *CombinedTM* class from the *contextualized_topic_models* library with specific optimizations for the Vietnamese corpus. We again employed the *paraphrase-multilingual-MiniLM-L12-v2* transformer to generate the contextual input (e), selected for its robust performance in multilingual semantic similarity tasks. To enforce the sparsity characteristics typical of Dirichlet priors found in statistical models, we utilized the **softplus**

activation function in the hidden layers; this constrains outputs to non-negative values, promoting sharper and more distinct topic definitions (Srivastava & Sutton, 2017). Furthermore, acknowledging that neural variational models are inherently stochastic, we implemented a rigorous **deterministic seeding strategy** across all computational frameworks (Python, NumPy, and PyTorch). This ensures that the inference process is stable and that the experimental results are scientifically reproducible.

3.7. Evaluation Framework

Evaluating unsupervised topic models is notoriously difficult due to the lack of "ground truth" labels. To objectively benchmark the performance of the five architectures, we utilized two quantitative metrics that act as proxies for human interpretability.

Topic Coherence measures whether the top words in a topic are semantically related. We specifically employed the C_v measure, which has been shown to have the highest correlation with human judgment (Röder, Both, & Hinneburg, 2015). Mathematically, C_v uses a sliding window over the corpus to calculate the Normalized Pointwise Mutual Information (NPMI) between the top-N words of a topic. A high C_v score (typically > 0.5) indicates that the words in a topic frequently co-occur and form a coherent concept (e.g., soldier, gun, mother), while a low score indicates a random mix of unrelated words.

High coherence is insufficient if a model produces redundant topics (e.g., five different topics all about "Love"). To penalize redundancy, we measured **Topic Diversity**. This metric is calculated as the percentage of unique words in the top-25 words of all topics (Dieng, Ruiz, & Blei, 2020).

$$Diversity = \frac{Unique\ Words}{Total\ Top\ Words}$$

A diversity score closer to 1.0 indicates that the model has successfully captured a wide range of distinct themes across the Vietnamese musical landscape, rather than collapsing into a few generic categories.

The selection of the "Champion Model" is not based solely on the highest numerical score. We apply a holistic selection criterion: the optimal model must achieve high Coherence and Diversity while demonstrating the qualitative ability to distinctly separate "Traditional" vocabularies from "Modern" ones. It must detect specific cultural nuances, such as distinguishing the specific festivity of *Tết* from the general season of *Spring*.

CHAPTER 4: EXPERIMENT AND RESULTS

4.1. Experimental Data

Following the rigorous preprocessing pipeline described in Chapter 3, the final corpus consists of **61,426 valid songs**. The filtering process—specifically the removal of high-frequency stopwords and the pruning of words appearing in more than 40% of documents—resulted in a highly condensed dataset focused on thematic content rather than grammatical structure. As summarized in Table 4.1, the final vocabulary size is reduced to **7,884 unique compound tokens**, representing the "semantic core" of the Vietnamese musical language. The average song length is **41.2 tokens**, reflecting the succinct nature of lyrical poetry after non-essential words are removed.

Metric	Value	Description
Total Valid Songs	**61,426**	Final count after cleaning and deduping
Total Tokens	2,533,746	Total semantic units processed
Vocabulary Size	7,884	Unique compound tokens (after pruning)
Average Length	41.2 tokens	Mean semantic density per song
Median Length	34.0 tokens	Robust measure of central tendency
Max Length	1,032 tokens	Outlier (Narrative poetry/Epic)
Missing Years	58.7%	Percentage of songs without verified release date

Table 4.1: Summary of Statistical Characteristics of the Corpus

The preprocessing stage demonstrated high efficiency in noise reduction. As detailed in our rejection statistics, the system removed approximately 8.7 million stopword tokens and over 291,000 Part-of-Speech (POS) artifacts. This confirms that the raw lyrics were heavily saturated with grammatical particles (e.g., là, và, những) and generic pronouns, which would have otherwise clouded the topic models.

As illustrated in Figure 4.1 below, the document length distribution follows a long-tail pattern. While the majority of songs are concise (20-60 tokens), the dataset retains significant outliers such as *Minh họa Kiều* (1,032 tokens) and *Thi Ca Lịch Sử Cuộc Đời Đức Phật Thích Ca* (839 tokens). These outliers typically represent narrative poetry or religious recitations and were preserved for their rich semantic value.

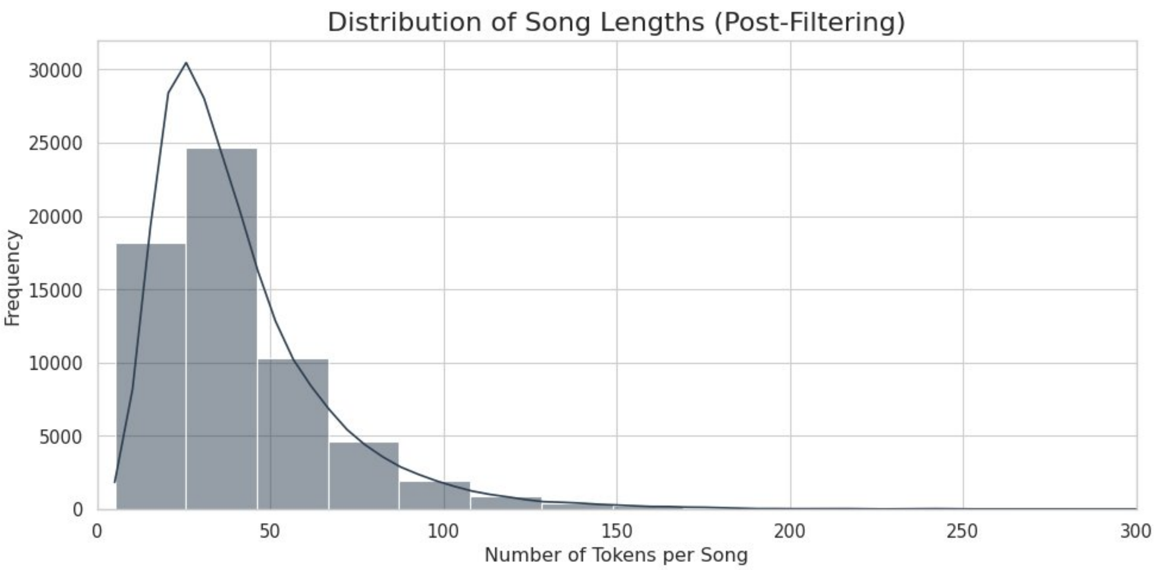

Figure 4.1: Distribution of song lengths after filtering

A critical challenge in reconstructing the “Quantitative History” of Vietnamese music is the scarcity of verified temporal metadata. Despite the implementation of API enrichment pipelines, **58.7% (36,070 songs)** of the dataset remains without a verified release year. The remaining verified subset of **25,356 songs** exhibits a distinct “Recency Bias,” as shown in Figure 4.2. The distribution is heavily skewed toward the post-2010 digital era (V-Pop), reflecting the ease of data collection for modern music compared to the archival scarcity of Pre-1975 recordings. Consequently, the **Historical Analysis (Chapter 5)** will be strictly limited to this verified subset to ensure chronological accuracy, while the general **Model Benchmarking (Section 4.3)** utilizes the full dataset to maximize semantic density.

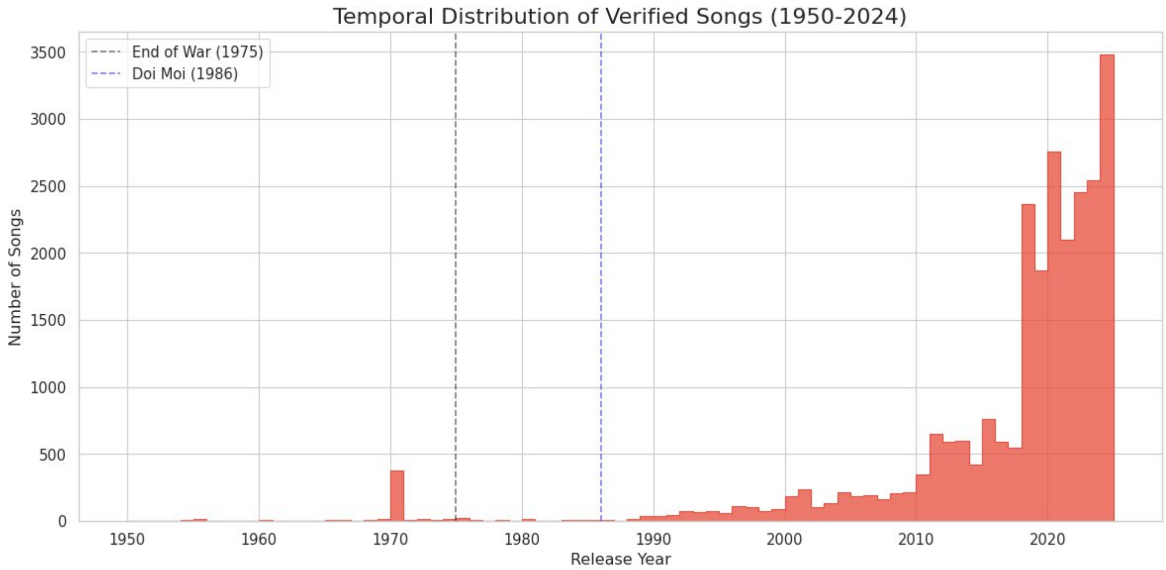

Figure 4.2: Temporal distribution of the verified subset (1950–2025)

An analysis of the most frequent tokens provides an immediate insight into the emotional landscape of the corpus. The lexicon is not neutral; it is dominated by emotive and natural imagery as seen in Figure 4.3. The high frequency of words such as *quên* (forget), *buồn* (sad), *đau* (pain), and *khóc* (cry) statistically confirms the hypothesis that Vietnamese popular music has historically favored melancholic and introspective themes. Furthermore, nature serves as the primary vehicle for this emotional expression, with *mưa* (rain), *gió* (wind), and *hoa* (flower) appearing in the top 15 most frequent tokens. Notably, the presence of *mẹ* (mother) in the top 10 underscores the enduring importance of filial piety in Vietnamese culture, a distinct trait compared to Western pop corpora where romantic love often dominates exclusively.

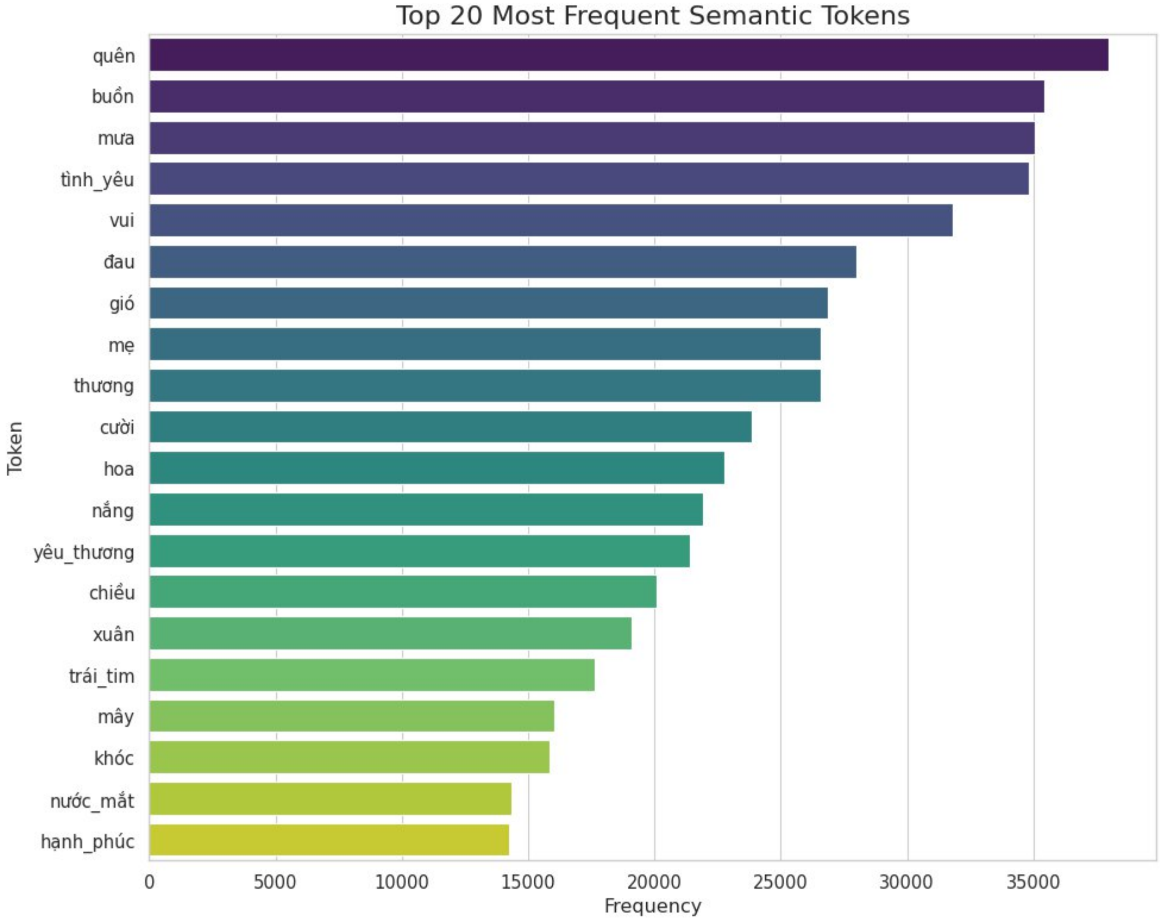

Figure 4.3: Top 20 most frequent semantic tokens in the corpus

4.2. Experimental Setup

The experimental framework was implemented on the **Google Colab** cloud platform. It is important to note that while the entire codebase is theoretically compatible with standard CPU environments, the training of transformer-based architectures imposes significant computational overhead. Specifically, the embedding generation for **BERTopic** and the neural variational inference for **Contextualized Topic Models (CombinedTM)** result in prohibitive training latency on CPUs. Therefore, a **GPU-accelerated environment** is strongly recommended and was utilized for this study. We employed an **NVIDIA T4 Tensor Core GPU** to handle these intensive neural tasks, reducing training time from hours to minutes and ensuring experimental feasibility.

We implemented a consistent Grid Search strategy for all unsupervised models to determine the optimal number of topics (K). We tested topic counts ranging from **K=6** to **K=31** with a **step size of 2**. This specific range was selected to identify the “Goldilocks zone”—the theoretical point where the model captures enough semantic detail to be meaningful without fragmenting the topics into unintelligible micro-clusters.

4.3. Results

Following the experimental grid search procedure outlined in Section 4.2, we evaluated the performance of all five models across various topic counts.

4.3.1. Latent Dirichlet Allocation (LDA)

Our quantitative analysis identified a clear performance peak at **K=12** with a **Coherence Score of 0.5646**. Beyond this point, increasing K resulted in diminishing returns, leading to fragmented clusters and lower semantic clarity.

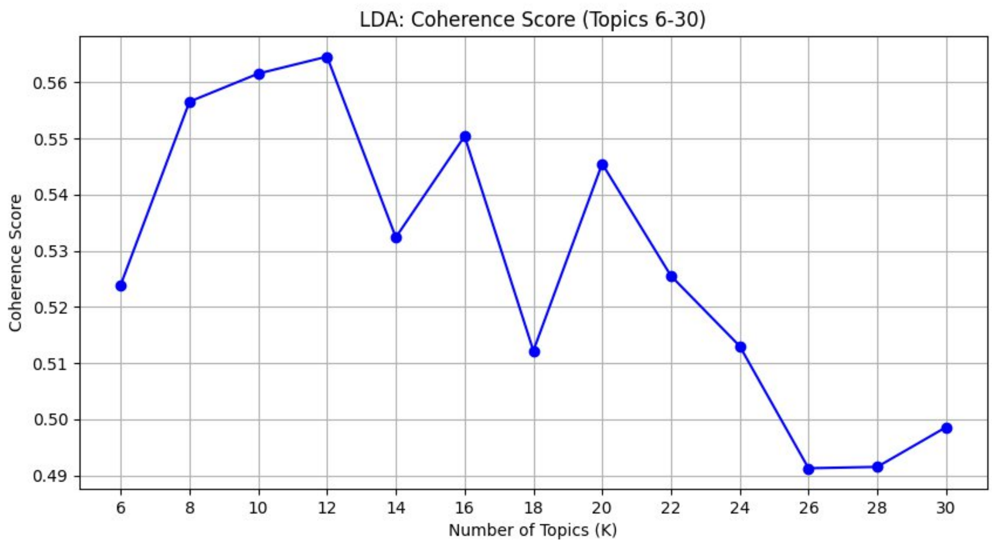

Figure 4.4: Coherence Score for LDA model

The resulting 12-topic model successfully disentangled the major cultural narratives of the corpus, achieving a **Topic Diversity Score of 0.7267**. This score is significantly higher than linear baselines like LSA (0.6200), indicating that the asymmetric alpha prior successfully forced the model to minimize redundancy.

The qualitative analysis of the LDA clusters reveals a sophisticated separation of Vietnamese cultural domains:

- **Topic 4 (Patriotism/Revolution):** A highly coherent "Red Music" cluster defined by keywords such as *quê_hương, việt_nam, máu, lửa, đất_nước* (Homeland, Vietnam, Blood, Fire, Country). This effectively captures the vocabulary of the Resistance era.
- **Topic 2 (Spring & Nature):** A seasonal cluster highlighted by *xuân, hoa, nắng, gió, chim* (Spring, Flower, Sun, Wind, Bird).
- **Topic 11 (Family & Childhood):** A synthesized cluster reflecting traditional values, containing *mẹ, cha, ru, hiền, cơm, tuổi_thơ* (Mother, Father, Lullaby, Gentle, Rice, Childhood).
- **Topic 8 (Belief & Religion):** A distinct spiritual topic containing *nguyện, thầy, phật, lạy, trần_gian* (Pray, Master, Buddha, Bow, Earthly World).

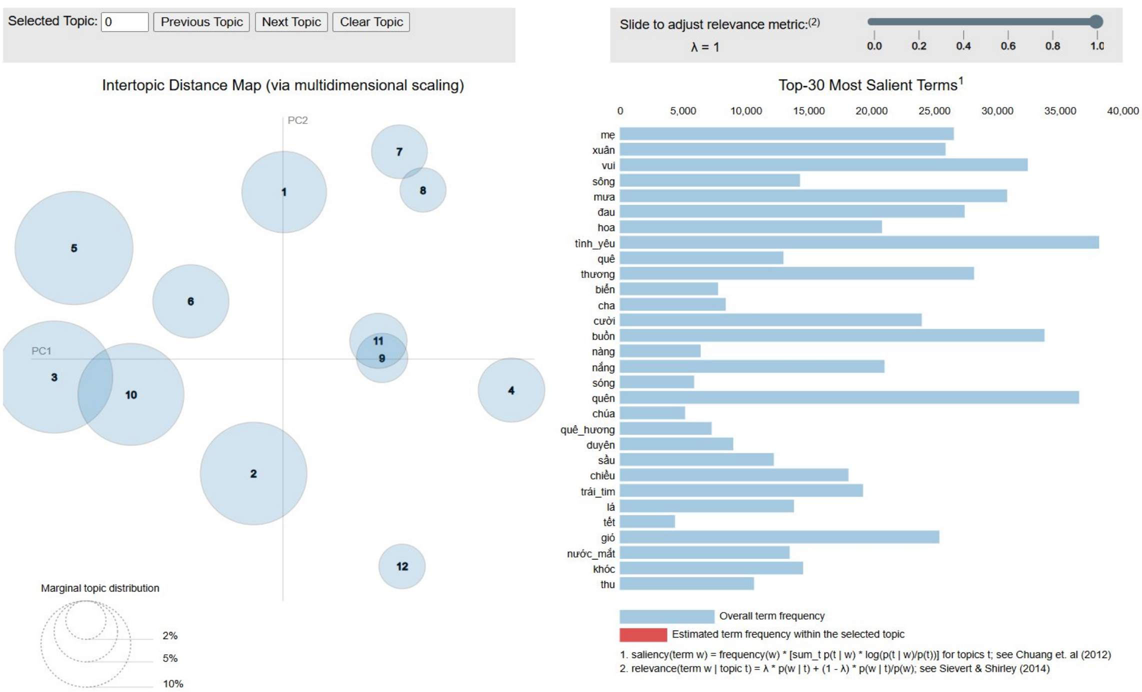

Figure 4.5: Inter-topic Distance Map generated by pyLDAvis

4.3.2. Non-negative Matrix Factorization (NMF)

Consistent with our rigorous experimental design, we replicated the Grid Search methodology used in the LDA stage.

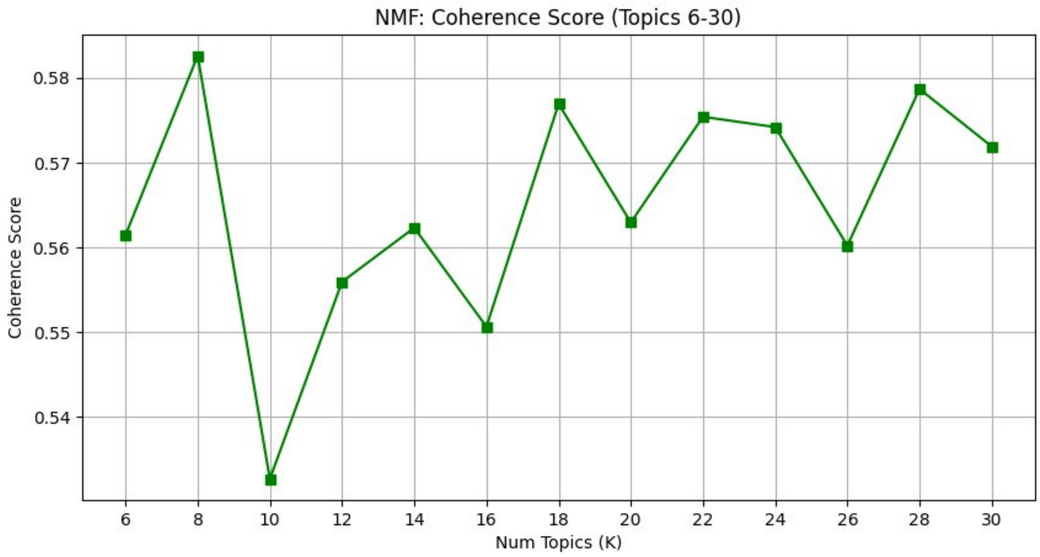

Figure 4.6: Coherence Score for NMF model

The quantitative results revealed a distinct divergence from the probabilistic baseline. While LDA peaked at 12 topics, the NMF model achieved optimal coherence at **K=8** with a superior **Coherence Score of 0.5826**. Furthermore, it achieved a **Topic Diversity Score of 0.7000**. This suggests that from a strict linear decomposition perspective, the Vietnamese musical corpus can be efficiently compressed into eight core thematic pillars.

To interpret these sharper themes, we generated a custom visualization (topic_barchart.html) displaying the top weighted terms:

- **Topic 3 (Spring & Celebration):** Unlike LDA's broader approach, NMF isolated a specific festive cluster defined by *xuân, tết, mai, đào, cười* (Spring, Tet, Ochna, Peach Blossom, Laugh).

- **Topic 7 (Filial Piety):** A highly coherent cluster containing *mẹ, cha, ru, hiền, lớn_khôn* (Mother, Father, Lullaby, Gentle, Grow Up), providing a distinct counterpoint to the romantic themes.
- **Topic 5 (Countryside & Romance):** A cluster focused on rural relationships, defined by *sông, quê, bến, chồng, duyên* (River, Countryside, Wharf, Husband, Fate).

Figure 4.7: Top weighted words for each NMF topic

4.3.3. BERTopic (Neural Topic Modeling)

The model achieved a **Coherence Score of 0.3991**. While this raw score appears lower than the probabilistic models, this is a known characteristic of embedding-based models which optimize for semantic density rather than word co-occurrence. However, the defining metric for BERTopic was its **Topic Diversity Score of 0.7357** at **K=28**.

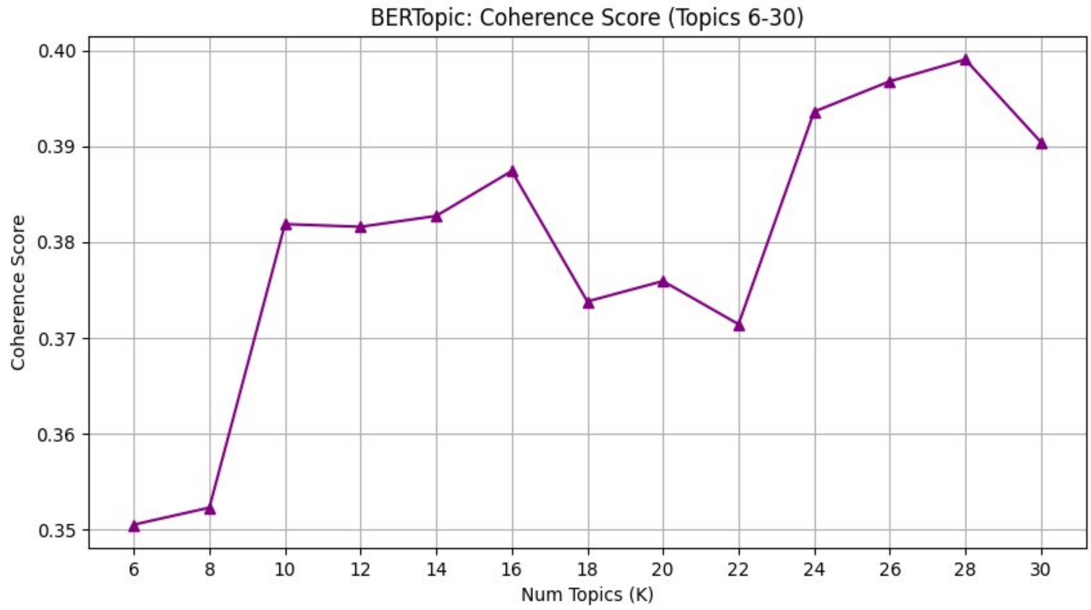

Figure 4.8: Coherence Score for BERTopic model

The true strength of BERTopic was revealed in its qualitative output. Unlike LDA/NMF, BERTopic detected highly specific, modern, and niche sub-genres:

- **Topic 8 (Digital Life):** A uniquely modern cluster containing *facebook, mạng, đăng, bạn_bè* (Facebook, Internet, Post, Friends), capturing the digital existence of the younger generation.
- **Topic 16 (Money & Materialism):** A distinct cluster characterized by *tiền_bạc, money, ngân, đồng_tiền* (Money, Bank, Currency), reflecting modern societal concerns.
- **Topic 20 (Christmas):** While other models found general "Festive" topics, BERTopic isolated "Christmas" with high precision: *christmas, giáng_sinh, nhiệm_màu* (Christmas, Noel, Miraculous).
- **Topic 17/18 (English Loanwords):** BERTopic successfully clustered songs containing English/Vietnamese code-switching, identifying terms like *lonely, goodbye, happy_birthday*.

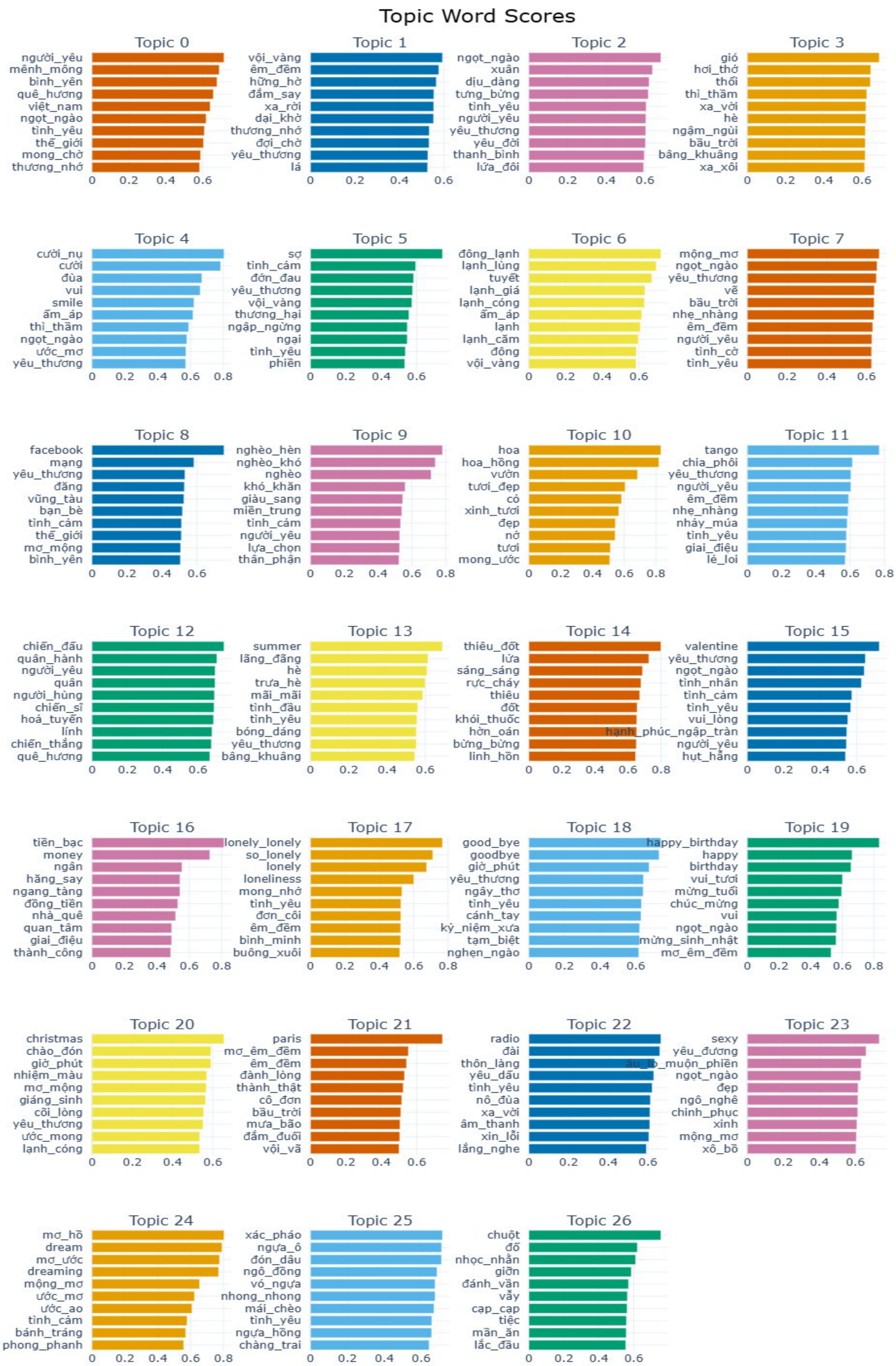

Figure 4.9: Top keywords for BERTopic clusters using KeyBERT representation

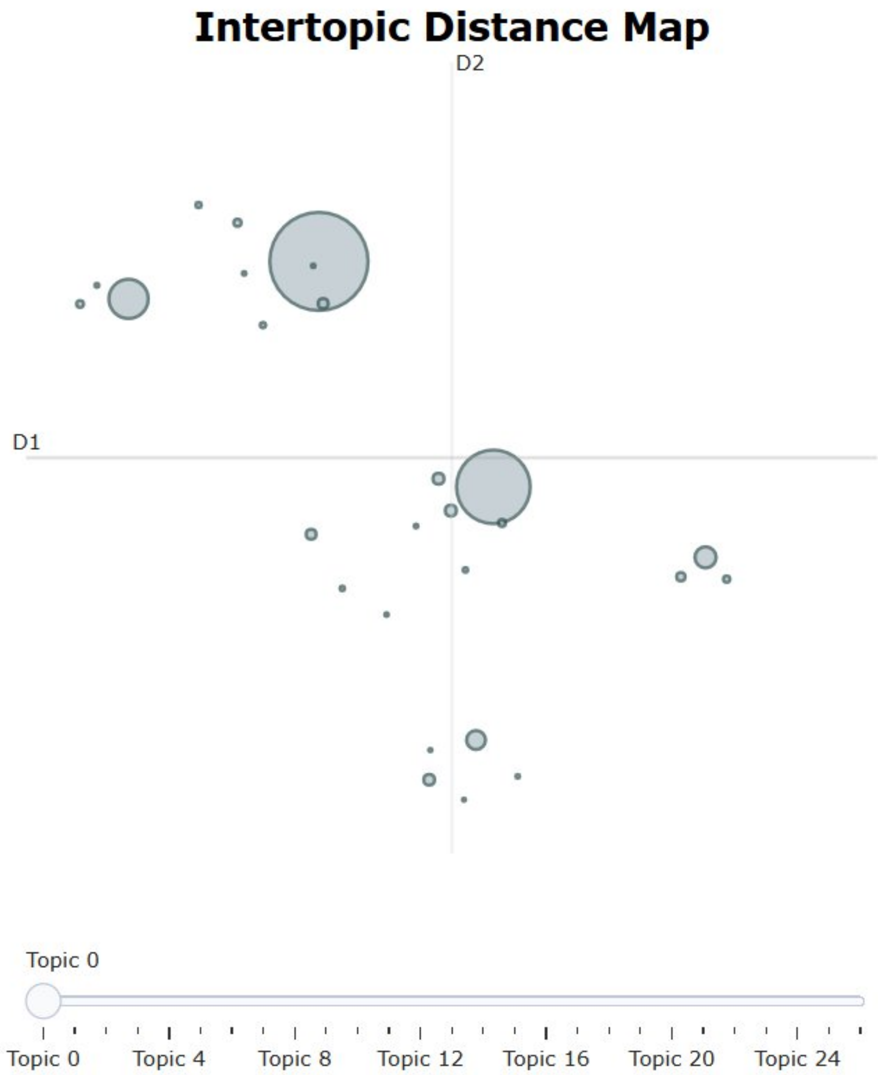

Figure 4.10: Interactive distance map of BERTopic clusters

4.3.4. Latent Semantic Analysis (LSA)

The model achieved a **Coherence Score of 0.5651** at **K=6**. This suggests that by rigorously removing the top frequent words, we effectively mitigated the noise that usually hampers linear models. However, the **Topic Diversity Score of 0.6200** remains the lowest among all models, indicating significant semantic overlap.

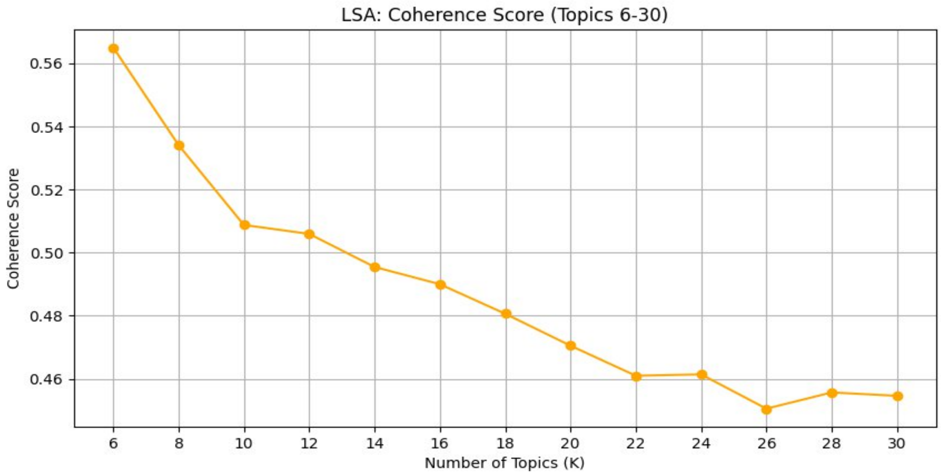

Figure 4.11: Coherence Score for LSA model

Qualitative analysis reveals "semantic bleeding" in the components:

- **Topic 1 (The "Mixed" Component):** As is common with LSA, the first component acts as a "super-topic," blending unrelated concepts like *mẹ* (Mother), *xuân* (Spring), and *khóc* (Crying).
- **Topic 4 (Nature & Melancholy):** A cluster aggregating standard poetic imagery: *thu, lá, sông, trăng, thuyền* (Autumn, Leaf, River, Moon, Boat).
- **Topic 6 (Separation):** Focused on negative sentiment with words like *khóc, chia_tay, trách, chết* (Cry, Breakup, Blame, Die).

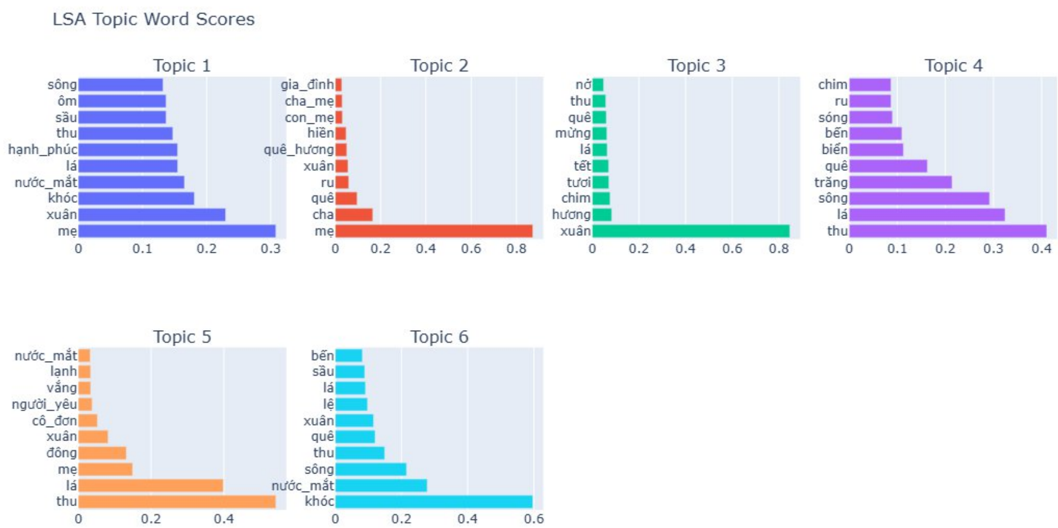

Figure 4.12: Top weighted words for LSA topics

4.3.5. Contextualized Topic Models (CTM)

The CTM model achieved a **Coherence Score of 0.6385** at **K=20**, which is significantly higher than the standard LDA model (0.5646) and NMF (0.5826). This quantitative leap validates the theoretical premise of (Bianchi, Terragni, & Hovy, 2021): incorporating contextualized representations allows the model to find topics that are more semantically "tight." Furthermore, the **Topic Diversity Score of 0.7740** is the highest in our experiment, indicating minimal redundancy.

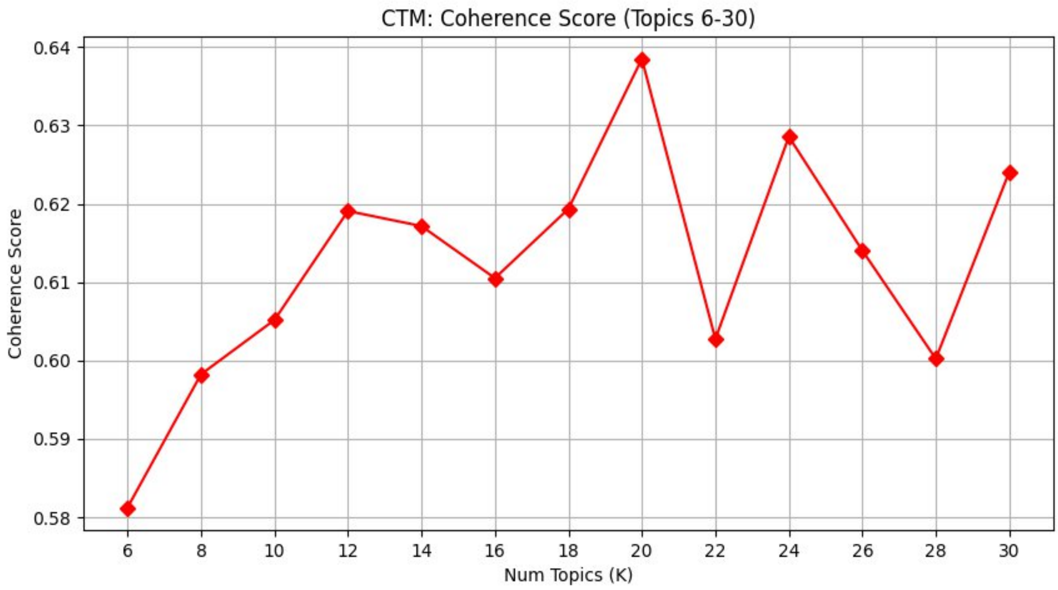

Figure 4.13: Coherence Score for CTM model

Qualitative analysis of the 20 topics reveals the most sophisticated cultural segmentation of the study:

- **Topic 12 (Modern Youth/Gen Z):** A clear departure from traditional vocabulary, containing slang and modern concepts like *rap, flow, baby, chill, điện_thoại* (Rap, Flow, Baby, Chill, Phone). Standard LDA failed to isolate this dialect.
- **Topic 20 (School Nostalgia):** A highly specific topic focused on student life, defined by *học_trò, phượng, thầy_cô, lưu_bút, xe_đạp* (Student, Phoenix Flower, Teachers, Yearbook, Bicycle).

- **Topic 8 (Southern Folk Culture):** Uniquely identified by CTM, this topic captures the vocabulary of Southern Vietnam's folk music (*miền_tây, sáo, đò, lục_bình* - West, Flute, Boat, Water Hyacinth).
- **Topic 15 (Religious/Spiritual):** A dedicated cluster for religious music, distinct from general belief, containing *chúa, phật, nguyện, từ_bi, giáng_sinh* (God, Buddha, Pray, Compassion, Christmas).
- **Topic 3 (Heroic History):** Distinct from general patriotism, this topic focuses on the active elements of war: *súng, quân, chiến_thắng, anh_hùng* (Gun, Army, Victory, Hero).

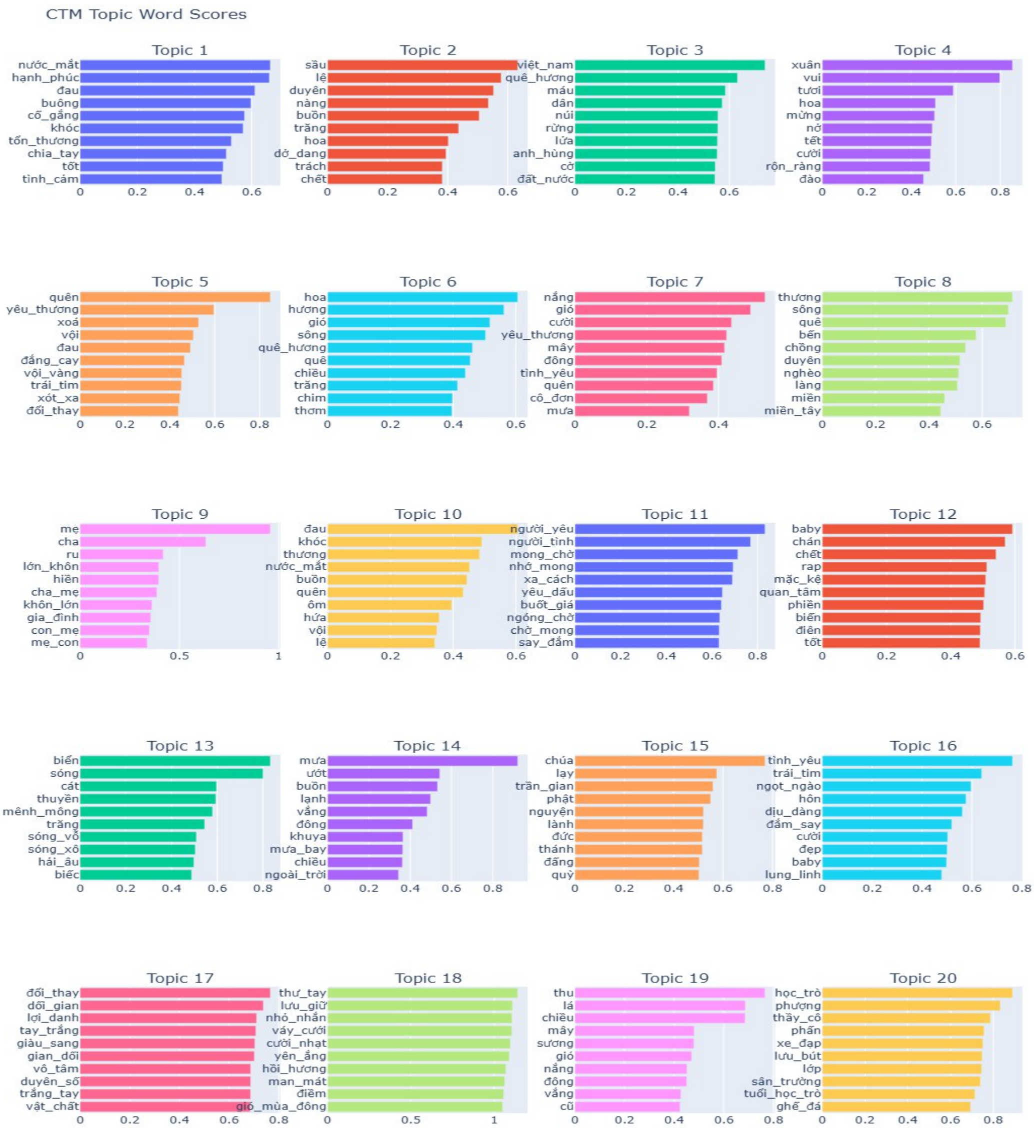

Figure 4.14: Top weighted words for CTM topics

4.3.6. Model Comparison and Selection

To identify the optimal methodology for this study, we aggregated the performance metrics of all five models derived from the experiments. Table 4.2 below presents a direct comparison of the optimal configurations for each model

Model	Optimal K	Coherence Score	Diversity Score	Qualitative Assessment
LSA	6	0.5651	0.6200	Fast but suffers from high semantic overlap; topics are too broad.
LDA	12	0.5646	0.7267	Solid baseline for traditional themes; struggles with slang and short lyrics.
NMF	8	0.5826	0.7000	Good at identifying specific events (Tet, Christmas) but lacks granularity (K=8).
BERT	28	0.3991	0.7357	Excellent for niche sub-genres (Digital Life) but low overall coherence; difficult to generalize.
CTM	**20**	**0.6385**	**0.7740**	**Superior performance.** Effectively combines semantic understanding with interpretable clusters.

Table 4.2: Performance Comparison of Topic Models

Based on the quantitative metrics and qualitative analysis, **CTM (Contextualized Topic Model)** is selected as the final model for this project because

CTM achieved the highest Coherence Score (**0.6385**), exceeding the standard LDA baseline by roughly 13%. It also achieved the highest Topic Diversity (**0.7740**), indicating that it produced the most distinct and non-redundant topics. Other than that CTM was the only model capable of distinguishing between “Traditional Romance” and “Modern Youth Culture” (identifying slang like *rap, flow, baby*), as well as correctly clustering “Southern Folk” music based on geographical vocabulary (*miền_tây, đò, sáo*).

To facilitate a structured historical analysis, the 20 latent topics identified by the CTM model were analyzed and manually labeled based on their top weighted keywords. As shown in Table 0.1 (can be found in the Appendix), these topics were categorized into six macro-thematic groups: *Love & Romance, Sadness & Mood, Homeland & Country, Tradition & Family, Modern Life & Society* and *School Days.*

CHAPTER 5: HISTORICAL ANALYSIS

5.1. Quantitative Historical Analysis

The core objective of this section is to perform a **Quantitative Historical Analysis** on the corpus of Vietnamese lyrics, a dataset characterized by its vast diversity in authorship, thematic content, and linguistic expression over the last century. It is crucial to emphasize that this approach is **descriptive and verify-driven** rather than predictive. By leveraging the scale of the dataset, which encompasses a significant portion of the recorded musical history, we aim to re-evaluate qualitative assertions previously made in the Social Sciences and Humanities—such as the "romantic nature" of pre-1945 music or the "political mobilization" of wartime compositions—through an objective, data-driven lens. Furthermore, the application of **Unsupervised Clustering** allows the analysis to transcend the subjective biases inherent in manual labeling (e.g., labeling a song as "cheesy" or "propaganda"), thereby revealing the latent, data-driven structures of Vietnam's cultural evolution.

5.1.1. Model Diagnostics and Validation

Before interpreting historical trends, it is methodologically necessary to distinguish between global model quality and local classification certainty. While **Topic Coherence** evaluates the semantic consistency of a topic's keywords, the **Confidence Score** represents the model's probability estimate that a specific song belongs to that topic.

A. Statistical Distribution and Lyrical Ambiguity

As illustrated in Figure 5.1, the distribution of confidence scores across the dataset reveals a mean probability of 0.29. In standard Natural Language Processing tasks, such as long-form news classification, confidence scores typically exceed 0.8. However, this lower threshold is consistent with the "Short Text Sparsity Problem" identified in computational linguistics literature. Research by (Hong & Davison,

2010) establishes that short texts—such as tweets or song lyrics—lack the rich statistical signals found in longer documents, leading to "sparse" feature vectors.

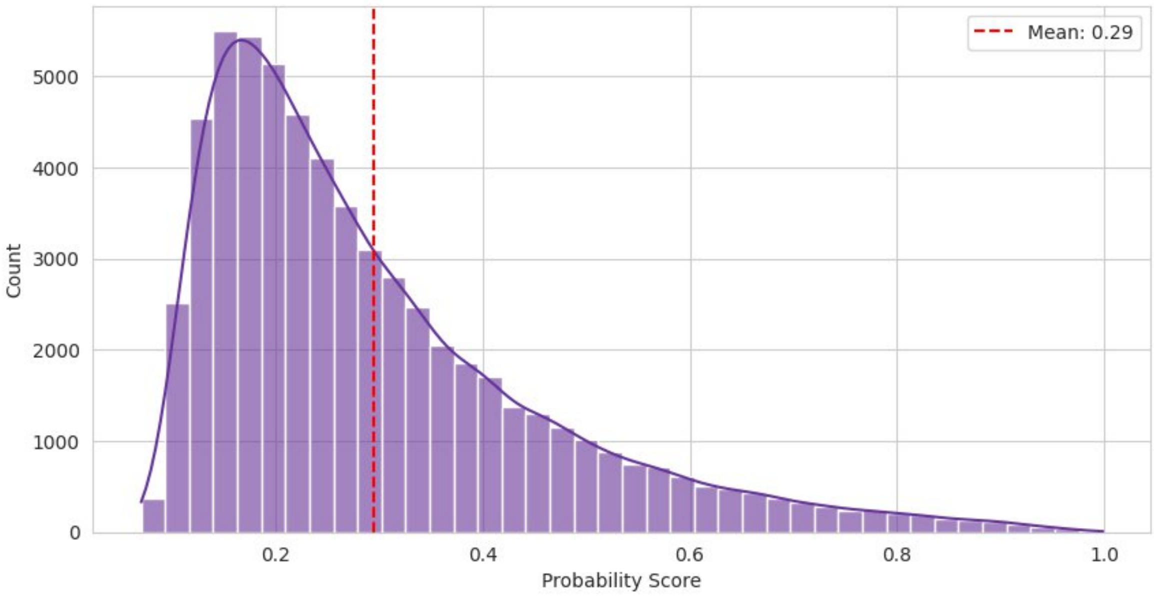

Figure 5.1: Distribution of Model Confidence

Furthermore, lyric poetry is inherently polysemous; a single song often acts as a **"semantic intersection"** where multiple themes collide. For example, a song about *Mother* (Family) may heavily utilize the vocabulary of *tears* (Sadness) and *river* (Homeland). Consequently, the model's lower confidence is not a failure, but an accurate reflection of the **high entropy** and poetic ambiguity inherent in the Vietnamese artistic corpus (Nguyen, Nguyen, Vu, Dras, & Johnson, 2018).

B. Topic Distinctiveness Analysis

While the average confidence is low, Figure 5.2 demonstrates that distinctiveness is highly dependent on the genre. The boxplot analysis reveals a clear hierarchy of interpretability. Topics such as "Religious Faith & Prayer" achieve the highest median confidence, as their lexicon (Chúa, Phật, nguyện) is domain-specific and rarely used in secular contexts. Conversely, "Waiting & Missing Lover" exhibits the lowest confidence and widest variance. This quantitatively confirms the "Background Radiation" hypothesis: romantic love vocabulary (yêu, nhớ, chờ) permeates almost every other topic in the Vietnamese corpus, making it mathematically difficult to isolate as a discrete cluster.

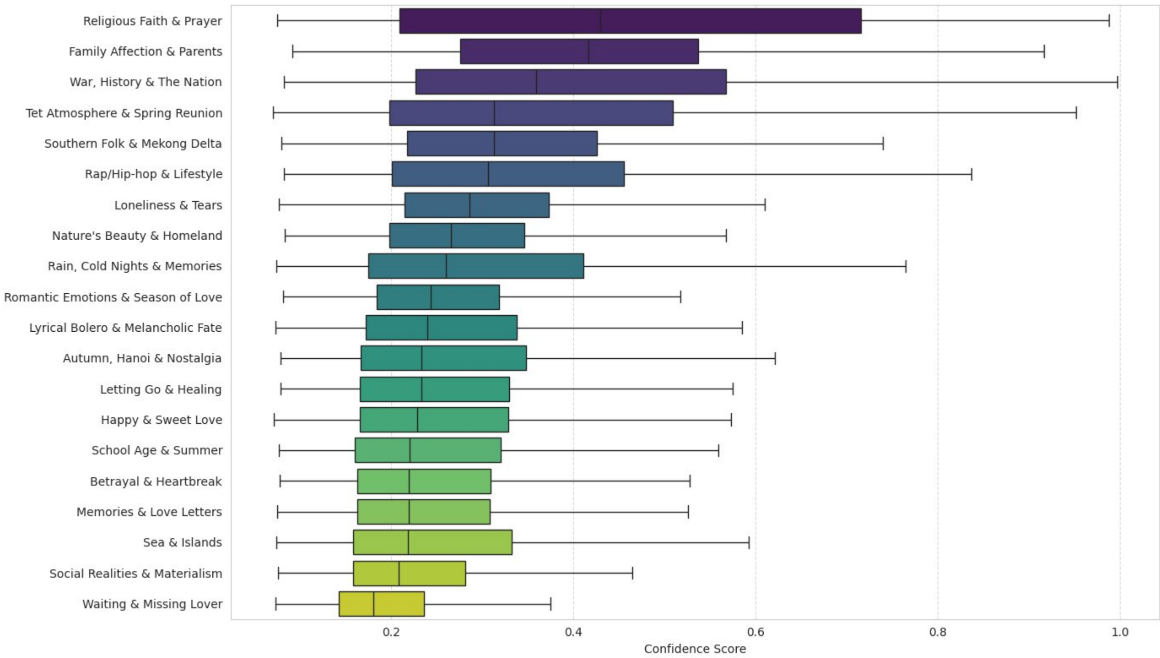

Figure 5.2: Distinctiveness of Each Music Genre

C. Authorial Validation (Composer Profiles)

To verify the semantic accuracy of the CTM labels, we cross-referenced the topics with their top-contributing composers using Figure 5.3. While the figure illustrates the model's alignment with real-world musical history across eight distinctive genres (including Bolero, Folk, and Materialism), for the sake of brevity, we highlight **4 representative examples** (Topics 0, 2, 11, and 19) that demonstrate the model's capacity to distinguish between linguistic nuance, historical context, and generational slang (please refer to Figure 0.1 in the Appendix for the complete composer profiles of all 20 topics):

- **Rap/Hip-hop (Topic 11):** The cluster is defined by the heavy presence of **Karik**, **B Ray**, and **Wrxdie**. The aggregation of these specific artists is significant because they represent the "genealogy" of Vietnamese Rap, spanning from its roots in the "Underground" scene to its current mainstream dominance. Their lyrics typically feature a higher density of syllables and distinct slang terminologies (*flow, hustling, chất*) compared to standard V-Pop. The fact that the model grouped these distinct generations of rappers together confirms its ability to isolate the unique linguistic "texture" of the Hip-hop genre, separating it from the melodic pop structures found in other topics.

- **Letting Go & Healing (Topic 0):** The cluster is dominated by **Mr. Siro**, an artist widely categorized by critics as the "King of Sad Ballads." His stylistic consistency in composing songs about heartbreak validates the model's ability to distinguish "Healing/Sorrow" from general "Love".
- **School Age & Summer (Topic 19):** The model's precision is further evidenced by the identification of **Nguyễn Ngọc Thiện** and **Vũ Hoàng** as the top contributors to this topic. In Vietnamese music history, these two composers are synonymous with the *Nhạc học đường* (Student Music) movement, famous for anthems regarding student life, summer farewells, and nostalgia. Their consistent use of specific imagery—such as *phượng vĩ* (phoenix flower), *sân trường* (schoolyard), and *lưu bút* (yearbook)—allowed the CTM model to carve out a niche, temporally-bound topic that is distinct from the broader "Youth Music" category.
- **War, History & The Nation (Topic 2):** The grouping of **Trịnh Công Sơn** and **Phạm Duy** within this cluster serves as a prime example of the model's reliance on lexical co-occurrence rather than ideological classification. Historically, these figures are distinct from the standard "Revolutionary" (*Nhạc Đỏ*) genre; **Trịnh Công Sơn** is celebrated for his **Anti-war** (*Nhạc Phản chiến*) and romantic repertoire, while **Phạm Duy** is defined by an eclectic range spanning **Folk** (*Dân ca*) and **Pre-war** (*Tiền chiến*) music. However, their inclusion here is mathematically justified by the **shared wartime vocabulary** of the 1954–1975 era. Because their lyrics extensively utilize the same high-density keywords found in revolutionary tracks the model clusters

them together. This reflects the collective linguistic atmosphere of a nation at war, where composers across the political and artistic spectrum shared a common language of pain and patriotism, despite their differing perspectives.

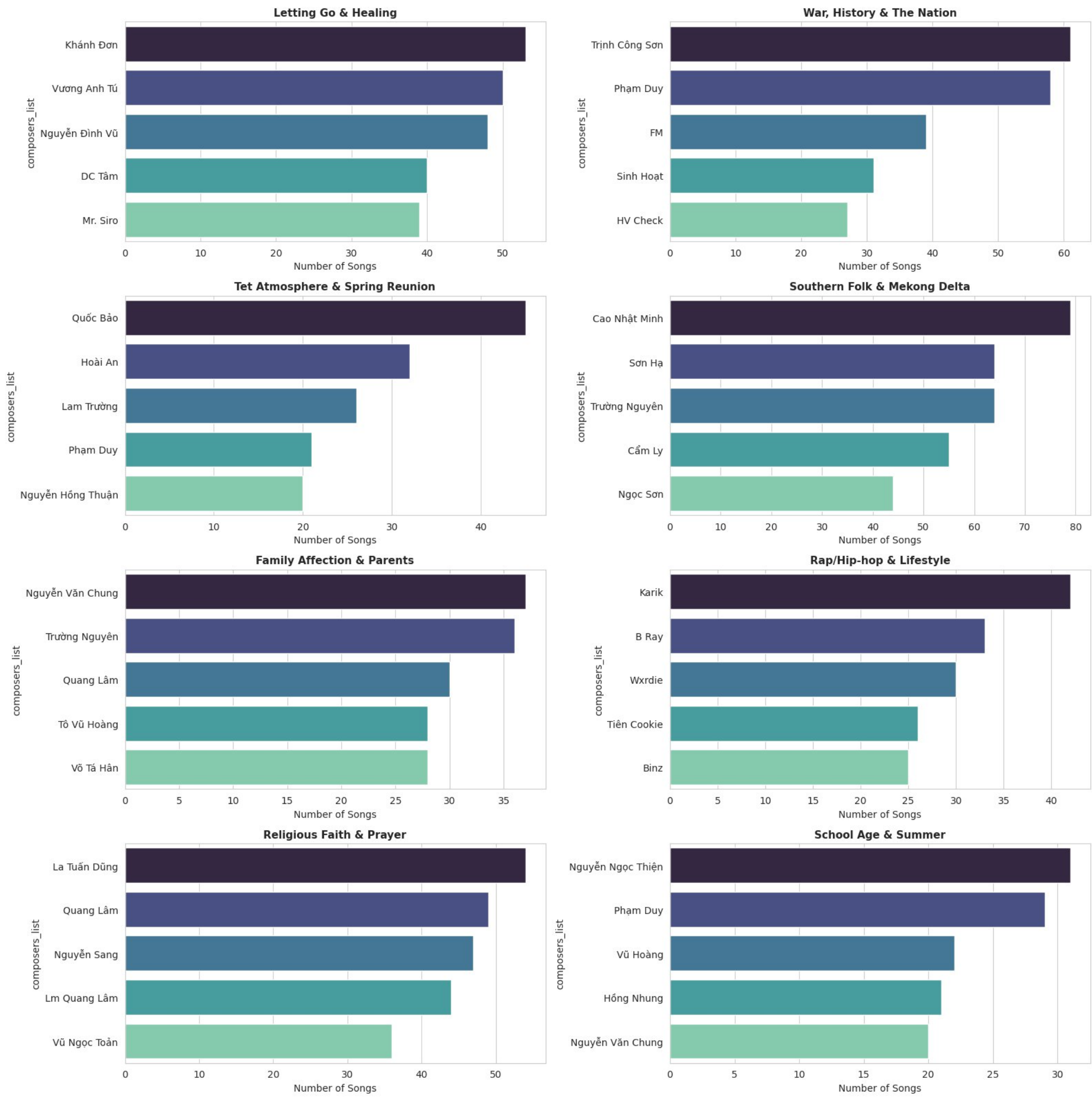

Figure 5.3: Top Composers in Distinctive Genre (topic 0, 1, 2, 3, 7, 11, 16, 19)

5.1.2. Historical Evolution of Vietnamese Music

The application of the Contextualized Topic Model (CTM) to the temporal dimension of the dataset reveals a profound transformation in the Vietnamese cultural psyche, tracing the nation's trajectory from **"Collective Survival"** to **"Individual Expression"**.

A. Macro-Analysis: The Three Eras of Vietnamese Music

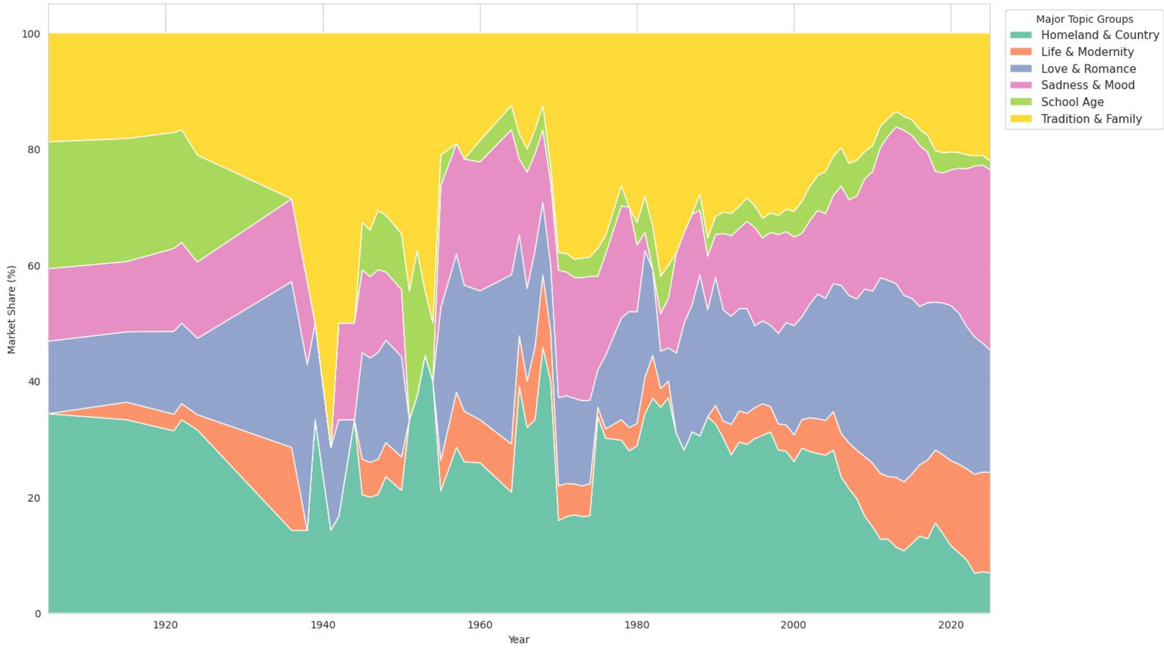

Figure 5.4: Evolution by Macro Groups

Figure 5.4 demarcates three distinct historical epochs. Chronologically, the first phase is **the "Resistance" Era (1940–1975)**, dominated by the **Homeland & Nation** group (Orange band), which accounted for nearly 50% of musical output during the Indochina Wars. This aligns with the historical consensus that music was mobilized as a "weapon of the spirit" to unify the populace.

Subsequently, the **"Subsidy" Transition (1976–1990):** The chart shows a contraction of political themes and an expansion of **Tradition & Family**, reflecting the introspection of the *Thời bao cấp* period. Economic hardship drove listeners toward family values and the comforting melancholy of Bolero.

Finally, the **"Individualist" Explosion (1995–Present):** The most dramatic structural break occurs post-Doi Moi. The "Homeland" theme collapses to a minority, replaced by a massive expansion of **Sadness & Mood** (Pink) and **Love & Romance** (Green). By 2025, these emotion-centric groups dominate over 60% of the market, statistically confirming the sociological shift toward individualism where personal emotional experience takes precedence over collective narratives.

B. Micro-Analysis: Cultural Flashpoints and Genre Trends

While the macro-view delineates broad historical currents, the **Small Multiples Analysis** (Figure 5.5) facilitates a granular examination of specific cultural flashpoints. By restricting the focus to a curated selection of eight representative genres—Topics 0, 1, 2, 3, 7, 11, 16, and 19—this analysis reveals how Vietnamese music simultaneously mirrors global evolutionary patterns while retaining unique local signatures (For a comprehensive visualization of the remaining topics, please refer to Figure 0.2 in the Appendix).

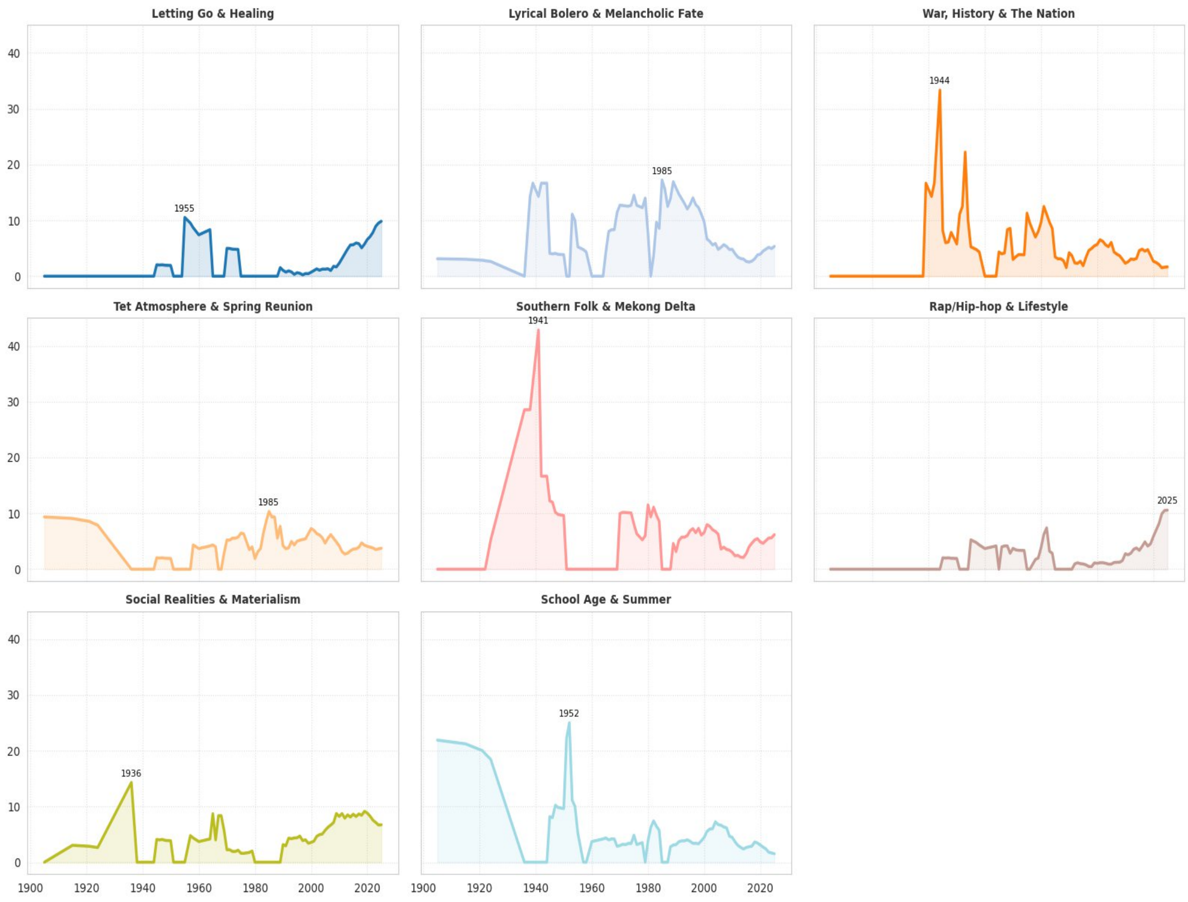

Figure 5.5: Small Multiples Analysis (topic 0, 1, 2, 3, 7, 11, 16, 19)

The "Sadness" Paradox: The data reveals that "Sadness" accounts for the largest volume of modern songs, raising the question of why this sentiment dominates. Domestic media attributes this to the "Prolactin Hypothesis", suggesting that listening to sad music triggers a hormonal release that soothes mental pain and creates a sense of peace rather than depression (VietnamPlus, 2019). Paradoxically, people often enjoy sad music even when they are not experiencing grief, as it acts as a stress relief mechanism that creates an emotional connection and gains the listener's empathy (VietnamNet, 2019). However, the temporal data indicates a qualitative shift: Vietnamese sadness has evolved from the fatalistic, resigned sorrow of the 20th century to the active "Letting Go & Healing" (Topic 0) of the 2020s. This transition to keywords like buông (let go) and chữa lành (heal) reflects a modern psychological approach to trauma, acting as a coping mechanism for a generation facing the pressures of modern urban life.

The "Rap" Synchronization: The most violent disruption in the dataset is Topic 11 (Rap/Hip-hop), which spikes vertically from 2018–2025. While this mirrors the global rise of Hip-hop culture emphasizing wealth and status, the local catalyst was specific. The year 2020 marked the "natural time, geographical advantage" (thiên thời, địa lợi) for the genre, coinciding with the broadcast of reality shows like Rap Viet and King of Rap, which brought the genre from the underground to the mainstream (Dan Viet, 2020). This explosion introduced a new lexicon of "lifestyle" and personal affirmation (flow, money, chất) that parallels the rise of Topic 16 (Social Realities & Materialism). As noted by industry observers, this event completely changed the market situation, proving that Rap is no longer a fad but a new pillar of the Vietnamese music market (Sai Gon Giai Phong, 2020).

Bolero as Cultural Escapism: Contradicting the decline of revolutionary music, Topic 1 (Lyrical Bolero) and Topic 7 (Southern Folk) display remarkable resilience. Bolero established itself as a "specialty" of the South in the 1950s, characterized by slow tempos and lyrics that tell the "everyday stories" (tâm sự) of the working class. The model captures a massive resurgence of this topic peaking around 1985, coinciding with the difficult Subsidy Era. Much like listeners elsewhere turn to idealized lyrics to escape reality, Vietnamese listeners used the fatalistic world of Bolero (sầu, lệ, phận) as a nostalgic refuge. The genre's ability to express the "heart of the masses" explains why it remains a core component of the Vietnamese identity, serving as a stable emotional anchor amidst rapid modernization (Tech Sound Vietnam, 2023).

Unique Local Signatures: While Rap and Sadness follow international patterns, Topic 2 (Revolutionary Spirit) and Topic 3 (Tet) remain uniquely Vietnamese. Topic 2's sharp spikes (1945, 1954, 1975) reflect the specific "mobilization" function of Vietnamese music, serving as a temporal fingerprint of the nation's conflicts. Furthermore, Topic 3 (Tet Atmosphere) displays a "heartbeat" pattern with regular annual spikes. Tet is considered the "golden time" for the music industry, where brands and artists release hundreds of songs annually to serve the shopping and reunion season (BaoMoi, 2021). The model's identification of stable keywords like xuân, mai, đào confirms that the cultural imperative of the Lunar New Year reunion remains the most constant, cyclic force in Vietnamese music history, impervious to the trends of globalization.

5.2. Discussion

The quantitative results of this study not only map the specific trajectory of Vietnamese music but also validate broader sociological theories regarding cultural evolution. By benchmarking our findings against international computational musicology studies, we can determine where Vietnam follows global trends and where it retains unique cultural resistance.

Vietnam in the Global Context: The "Sadness" Convergence Our analysis confirms that Vietnam aligns with the global trend identified by Schellenberg & von Scheve (2012), who documented a longitudinal shift in American pop music toward "sadder" and more emotionally complex content over five decades. Our corpus reveals a similar trajectory: the "Collective Optimism" of the Resistance Era (Topic 2) has been largely displaced by the "Individual Melancholy" of the modern era (Topic 0, 9). However, a critical divergence exists. While Western studies attribute this shift solely to consumer individualism, the Vietnamese context suggests a dual driver. The "Sadness Paradox" in our data (Section 5.1) indicates that modern Vietnamese sadness (Topic 0: *Letting Go & Healing*) functions as a therapeutic mechanism—a "coping strategy" for a generation navigating rapid urbanization—rather than purely aesthetic individualism. This aligns with the "Prolactin Hypothesis," suggesting that the Vietnamese market consumes sadness not to wallow, but to heal.

The "Doi Moi" Breakpoint vs. The "German Benchmark" Comparing our results to the benchmark study of German lyrics (Hunke, Huber, & Steffens, 2025), we observe striking parallels in post-conflict nations. Hunke et al. traced the decline of "Post-war longing" and the rise of "Status & Society" in Germany. Similarly, our CTM model captures the collapse of **Topic 2 (Revolutionary Spirit)** immediately following the end of the War (1975) and the subsequent explosion of **Topic 16 (Materialism)** and **Topic 11 (Rap/Lifestyle)** post-*Doi Moi*. This confirms that the economic liberalization of 1986 was not just a financial policy but a cultural "Event Horizon." It fundamentally altered the lyrical lexicon, shifting the grammatical focus from the collective "We" (*Ta/Non_sông*) to the possessive "I" (*Tôi/Tiền_bạc*), mirroring the trajectory of other post-socialist economies entering the global market.

Cultural Resistance: The Resilience of Tradition Despite these modernization trends, our study reveals a phenomenon of "Cultural Resistance" unique to Vietnam. While Mauch et al. (2015) argued that musical evolution is driven by "revolutions" that replace old genres, our data shows that certain traditional topics in Vietnam are immune to replacement. **Topic 3 (Tet Atmosphere)** and **Topic 1 (Lyrical Bolero)** do not decline over time but instead exhibit cyclical stability. The persistence of “Southern Folk” (Topic 7) and “Family/Parents” (Topic 8) challenges the assumption that globalization leads to cultural homogenization. Instead, Vietnamese music demonstrates a “dual-track” evolution: while the surface layer (production, slang) globalizes rapidly—as evidenced by the Rap spike—the core thematic layer (filial piety, lunar festivals) remains deeply entrenched, serving as an immutable anchor for national identity

CHAPTER 6: CONCLUSION

6.1. Conclusion

This study successfully applied **Quantitative Historical Analysis** to the vast, unstructured corpus of Vietnamese popular music, fulfilling its primary objective of re-evaluating qualitative historical narratives through an objective, data-driven lens. By leveraging the **Contextualized Topic Model (CTM)** on a dataset of over 60,000 songs, the research mapped the evolutionary trajectory of the Vietnamese cultural psyche, revealing a profound macro-structural shift from the **"Collective Survival"** of the wartime era to the **"Individual Expression"** of the modern age. The model accurately reconstructed the nation's history without supervision; the violent spikes of **Topic 2 (Revolutionary Spirit)** aligned precisely with the conflicts of 1945, 1954, and 1975, proving that music in the mid-20th century functioned primarily as a vehicle for national mobilization. Conversely, the analysis quantified the "Individualist Explosion" of the 21st century, where the exponential rise of **Rap/Materialism (Topic 11, 16)** and **Sadness (Topic 0, 4)** demonstrates that modern Vietnamese lyrics have integrated into global pop culture trends, prioritizing personal emotion, mental health awareness, and status seeking over collective narratives.

Despite rapid modernization, the study also revealed the deep-seated resilience of traditional values. **Topic 1 (Bolero)** and **Topic 3 (Tet Atmosphere)** emerged as cultural constants, serving as emotional anchors—one providing nostalgic escapism for the working class, the other marking the cyclic rhythm of family reunion impervious to global trends. Methodologically, the research demonstrated that while Vietnamese lyrics exhibit high linguistic entropy—evidenced by low confidence scores due to short-text sparsity—the **CTM architecture** successfully overcame this barrier. By cross-referencing topics with known composer profiles (e.g., Trịnh Công Sơn, Mr. Siro, Karik), we validated that the model could correctly identify genre-specific vocabularies, proving the viability of computational musicology for analyzing low-resource languages.

6.2. Limitations and Future Work

Despite the robust findings, this study is subject to specific methodological constraints, most notably the **“Short Text Sparsity” problem**. As analyzed in the diagnostics section, the average confidence score of the model was 0.29, consistent with literature on lyric mining. This indicates that a significant portion of the corpus contains “mixed membership” or ambiguous lyrics that defy strict categorization. Furthermore, the research focused exclusively on textual content, omitting audio features such as melody, tempo, and instrumentation. This limitation means that a “sad” lyric set to an upbeat tempo—a common trope in modern V-Pop—might be misclassified by a text-only model as purely melancholic. Additionally, while rigorous imputation was employed for release years, the reliance on estimated metadata for parts of the pre-1975 dataset may result in a slight smoothing of historical trends in earlier decades compared to the high-resolution data available for the digital era.

To build upon these findings, future research should integrate **Multimodal Analysis** by combining CTM topics with audio signal processing (valence, arousal, tempo) to distinguish between nuances like “sad ballads” and “sad dance tracks.” Following the approach of Hunke et al., future studies could also implement a specialized sentiment analysis model to track the specific evolution of “Ambiguity” and “Anger” in Vietnamese lyrics, testing whether the rise of Rap correlates with a rise in narcissistic vocabulary. Finally, given the vertical spike in “Rap/Lifestyle” topics post-2018, investigating the impact of platform algorithms (e.g., TikTok) is essential to determine if the need for “viral hooks” is driving a structural shift toward shorter, more repetitive, and materialistic lyrics in the Vietnamese market.

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APPENDIX

Group	ID	Topic Label	Count	Top Representative Keywords
Love & Romance	6	Romantic Emotions & Season of Love	2011	*nắng, gió, mây, đông, ôm, hôn*
	10	Waiting & Missing Lover	3771	*mong_chờ, ngóng, xa_cách, dĩ_vãng, tình_đầu*
	15	Happy & Sweet Love	3863	*ngọt_ngào, xinh, lung_linh, đắm_say*
	17	Memories & Love Letters	4706	*thư_tay, váy_cưới, lưu_giữ, man_mát*
Sadness & Mood	0	Letting Go & Healing	3178	*buông, chấp_nhận, kết_thúc, cố_gắng*
	4	Betrayal & Breakup	4258	*dối_gian, thay_đổi, xoá, đắng_cay, nát*
	9	Loneliness & Tears	2255	*khóc, lệ, cô_đơn, vỡ, một_mình*
	13	Rain, Cold Nights & Memories	3559	*mưa, ướt, lạnh, buốt_giá, khuya, lang_thang*
Homeland & Country	2	War, History & Nation	2474	*việt_nam, máu, súng, cờ, quân, anh_hùng*
	5	Nature's Beauty & Homeland	1722	*hoa, hương, suối, rừng, làng, sương*
	12	Sea & Islands	2928	*biển, sóng, hải_âu, thuyền, trùng_dương*
	18	Autumn, Hanoi & Nostalgia	3246	*thu, lá, hà_nội, đà_lạt, heo_may, cốm*
Tradition & Family	1	Lyrical Bolero & Melancholy Fate	3400	*sầu, lệ, duyên_kiếp, bến, đắng_cay*

	3	Tet Atmosphere & Spring Reunion	2940	*xuân, tết, đào, lộc, giao_thừa, gia_đình*
	7	Southern Folk & Mekong Delta	2957	*miền_tây, bến, sáo, lục_bình, phù_sa, chồng*
	8	Family Affection & Parents	2800	*mẹ, cha, công_ơn, tảo_tần, mồ_côi*
	14	Religious Faith & Prayer	2302	*chúa, phật, lạy, nguyện, cứu, thánh*
Modern Life & Society	11	Rap/Hip-hop & Lifestyle	2972	*rap, flow, baby, chán, tiền, ghét*
	16	Social Realities & Materialism	3816	*lợi_danh, giàu_sang, trắng_tay, bạc_tiền*
School Days	19	School Age & Summer	2268	*phượng, xe_đạp, thầy_cô, lưu_bút, sân_trường*

Table 0.1: Interpretative Labels for the CTM Model

Figure 0.1: Top Composers in Distinctive Genre (full 20 topics)

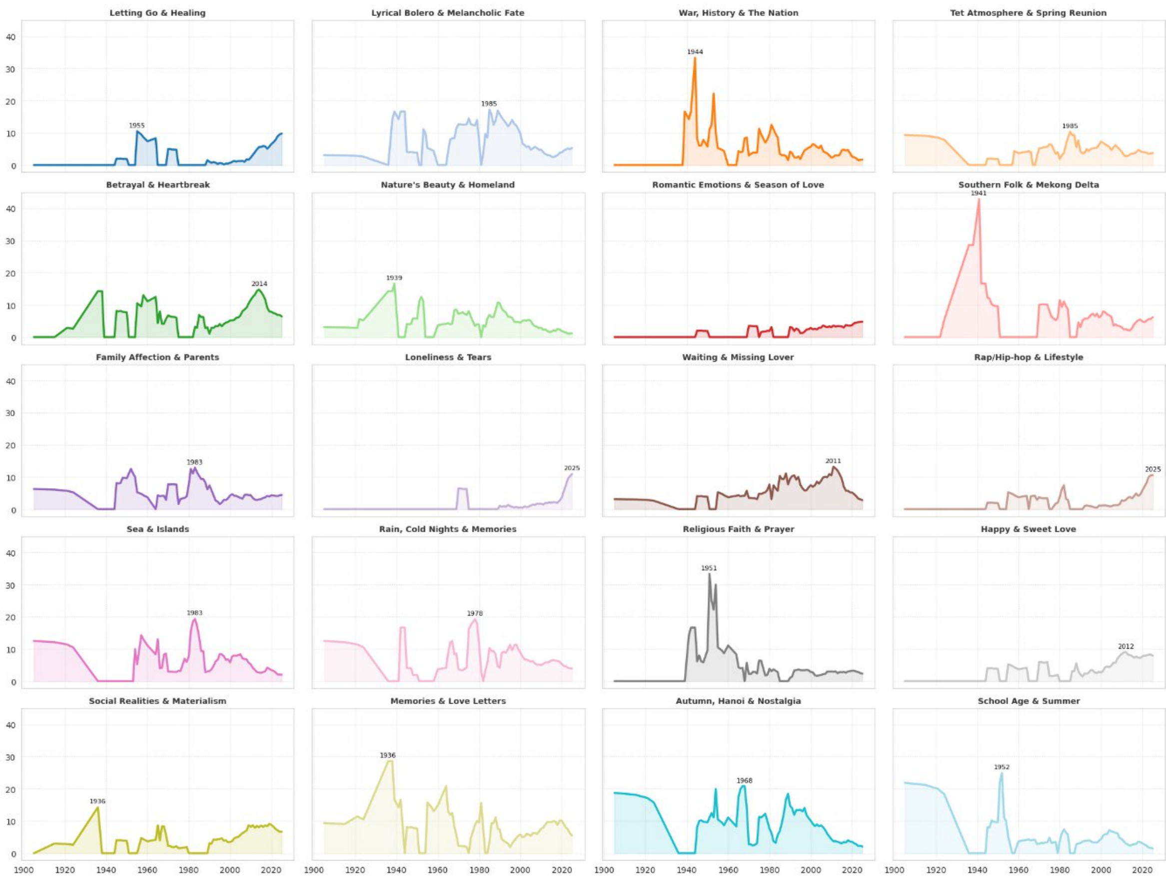

Figure 0.2: Small Multiples Analysis (full 20 topics)